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Lowest un-prescaled triggers per data-taking period

This twiki provides a short information on trigger chains recommended for physics analysis (non exhaustive list) which were un-prescaled through a major part of the data taking. More accurate information on pre-scales and parameters can be extracted from the trigger DB using the web browsers described below.

Please take note that pre-scales can change during various data taking periods. If a trigger chain was pre-scaled for a short period of time it still enters the table below. If lower thresholds are important for you, you can still use them, even if they are pre-scaled. Sometimes this would give you a better acceptance than using higher Pt un-prescaled triggers. Email hn-atlas-trigger-menus@cern.NOSPAMch or atlas-trigger-menu-expert@cern.NOSPAMch if you need help.

Getting information from the trigger DB

TrigConf portal (Run 2)

<https://atlas-trigconf.cern.ch/> - linked also from Run Query for Run 1, 2 runs - use "show all" or "show trigkeys" in run query and follow the links in the "Prescale keys" column (example "find r data18_13TeV.periodD / show trigkeys") for individual keys or for the prescale evolution.

TriggerToolWeb (Run 3)

<https://atlas-triggertool.web.cern.ch/> - linked also from Run Query for Run 3 runs - use "show all" or "show trigkeys" in run query and follow links in the "Prescale keys" column (disregard links for "prescale evolution" which incorrectly directs to a Run 2 page) (example "find r data22_13p6TeV.periodF / show trigkeys")

RunQuery

The Run Query (<http://atlas-runquery.cern.ch/>) is useful for getting the Super Master Key (SMK) as well as prescale keys and bunchgroup keys for a run with links to the TrigConf portal (Run 2) or TriggerToolWeb (Run 3), which contain further details (see above).

One can also:

- For Run 2, obtain link to run's prescale evolution in TrigConf Portal by querying a run/data period with "show all" or "show trigkeys" and clicking on "prescale evolution..." in "Prescale keys" column (example "find r data18_13TeV.periodD / show trigkeys").
- For Run 2 and 3, obtain a prescale evolution graph of L1/HLT triggers, presented as (x - - 2 x
2 - - - - - x) where x means disabled, - means no change, and numbers are the prescales for that prescale set (example "find r data22_13p6TeV.periodF / show tr HLT_j420*").
 - ◆ For L1 triggers, querying "trigrates " will also display a trigger rate graph (example "find r data22_13p6TeV.periodF / show tr L1_J100 and trigrates L1_J100").

(see the "Trigger" tab in the Run Query site for more features).

COMA

The COMA Project (Conditions/Configuration Metadata for ATLAS) is described in TWiki ConditionsMetadata. Its web-based interfaces can be found via the COMA Portal [COMA Portal](#). COMA is useful for quickly interrogating runs or periods, eg. for identifying the runs in each specific data period or sub-period, and interesting associated information. It also provides trigger and prescale related information by run or across many runs; useful to find which triggers went to which streams, which triggers were configured and/or active during runs or periods, and what were the prescales (or prescale ranges) during those runs. You will need the grid certificate uploaded to your browser to use the interface (see the TWiki page for help).

- The COMA Period Menu [COMA Period Menu](#) shows all ATLAS Data Periods. It offers links to reports describing each Period, its run/date range and other interesting information as well as links to other reports (AMI, Run Query, and COMA Reports).
- The COMA Report Menu (link from the COMA Portal page) allows user specification of a variety of input variables to generate COMA Reports: the COMA Report Type is determined by which input criteria is entered by the user (inputs include: Project, Period, Run, Stream Name, Trigger HLT Chain or L1 Item Name etc).

COMA Examples:

- Example **Quick Chain Query**
 - ◆ Enter Chain name: , then click:
- Example **GRL Chain Prescale Query**
 - ◆ Enter GRL name:
 - and Chain name , then click:

See the TWiki page for an overview of the COMA Reports and example links.

TriggerAPI

Update 18/06/2024: *The TriggerAPI has been updated for Run 3. Please try using it to obtain lists of unprescaled triggers of a given type for your analysis. The main input for the new API is a good runs list, and the API will tell you the unprescaled triggers for that GRL.

The Trigger API is a python interface to obtain information about the lowest-unprescaled triggers for a given signature and a given period. More information can be found on the dedicated TriggerAPI twiki page.

Triggers for low-mu runs

The list of lowest-unprescaled triggers for the 2017 and 2018 low-mu runs can be found here: [LowestUnprescaledLowMuRun2](#)

The list of lowest-unprescaled triggers for the 2022 low-mu runs will be filled in at [LowestUnprescaledLowMuRun3](#)

Triggers in 2025

Introduction

Recommended triggers for 2025 physics analyses can be found below. Known issues are detailed in [TriggerIssues2025](#).

See [TriggerNamingRun3](#) for naming conventions.

Trigger periods in intensity ramp-up (DRAFT)

Rulebook branches listed are the final tag used for that period, which contain the most stringent rules applied.

Intensity step	Data periods	First run number	Rulebook branch
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For runs in each data periods, see [COMA](#) [↗](#)

Egamma (DRAFT)

Trigger Group	Single electron	Multi electron
From 75 bunches	HLT_e26_lhtight_ivarloose_L1eEM26M HLT_e26_lhtight_ivarloose_L1eEM26T HLT_e20_lhtight_ivarloose_L1ZAFB-25DPHI-eEM18M HLT_e26_lhtight_ivarmedium_L1eEM26M HLT_e26_lhtight_ivartight_L1eEM26M HLT_e60_lhmedium_L1eEM26M HLT_e140_lhloose_L1eEM26M HLT_e140_lhloose_noringer_L1eEM26M HLT_e300_etcut_L1eEM26M Large-radius tracking HLT_e30_lhloose_nopix_lrtmedium_L1eEM26M	HLT_2e17_lhvloose_L12eEM18M HLT_2e24_lhvloose_L12eEM24L HLT_e24_lhvloose_2e12_lhvloose_L1eEM24L_3 HLT_e26_lhtight_e14_idperf_tight_probe_50inv HLT_e26_lhtight_e14_idperf_tight_nogsf_probe_50 HLT_e26_lhtight_e14_etcut_probe_50invmAB13

Trigger Group	Single photon	Multi photon
From 75 bunches	HLT_g140_loose_L1eEM26M HLT_g300_etcut_L1eEM26M	HLT_2g22_tight_L12eEM18M HLT_2g50_loose_L12eEM24L HLT_2g20_tight_icaloose_L12eEM18M HLT_g35_medium_g25_medium_L12eEM24L HLT_2g25_loose_g15_loose_L12eEM24L HLT_2g15_tight_25dphiAA_invmAA80_L1DPHI-M70-2eEM15M
From Run 476718		HLT_g45_medium_g20_medium_L1eEM40L_2eEM18L

Trigger Group	Electron + photon
From 75 bunches	HLT_e24_lhmedium_g25_medium_02dRAB_L12eEM24L HLT_e24_lhmedium_g12_loose_g12_loose_02dRAB_02dRAC_02dRBC_L1eEM24L_3eEM12L

Trigger Group	Electron + photon
	HLT_e25_mergedtight_g35_medium_90invmAB_02dRAB_L12eEM24L

Muon

Trigger Group	Single muon	Multi muon
75, 400 bunches only	N/A	N/A
From 900 bunches	HLT_mu24_ivarmedmedium_L1MU14FCH HLT_mu50_L1MU14FCH HLT_mu60_0eta105_msonly_L1MU14FCH HLT_mu60_L1MU14FCH HLT_mu80_msonly_3layersEC_L1MU14FCH HLT_mu3vtx_L12MU8F Large-radius tracking HLT_mu20_LRT_d0loose_L1MU14FCH	HLT_2mu14_L12MU8F HLT_mu20_ivarmedmedium_mu8noL1_L1MU14FCH HLT_mu22_mu8noL1_L1MU14FCH HLT_3mu6_L13MU5VF HLT_4mu4_L14MU3V HLT_2mu10_l2mt_L1MU10BOM Invariant mass cut HLT_mu10_ivarmedmedium_mu10_10invmAB70_L12MU8F HLT_mu20_ivarmedmedium_mu4noL1_10invmAB70_L1MU1

Electron/Photon + Muon (DRAFT)

Trigger Group	Electron + Muon	Photon + Muon
From 400 bunches	N/A	Photon + 2 muon HLT_g35_loose_mu15_mu2noL1_L1eEM28L
From 900 bunches	Electron + muon HLT_e7_lhmedium_L1eEM5_mu24_L1MU14FCH HLT_e17_lhloose_mu14_L1eEM18L_MU8F HLT_e26_lhmedium_mu8noL1_L1eEM26M 2 electron + muon HLT_2e12_lhloose_mu10_L12eEM10L_MU8F Electron + 2muon HLT_e9_lhvloose_L1eEM5_mu20_mu8noL1_L1MU14FCH HLT_e12_lhloose_L1eEM10L_2mu10_L12MU8F	Photon + muon HLT_g25_medium_L1eEM18L_mu24_L1MU14FCH HLT_g25_medium_L1eEM26M_mu24_ivarmedmedium_L1MU14FCH HLT_g35_loose_mu18_L1eEM28M HLT_g35_loose_mu15_mu2noL1_L1eEM28L HLT_g40_loose_L1eEM24L_mu40_msonly_L1MU14FCH 2 photon + muon HLT_2g10_loose_L1eEM9_mu20_L1MU14FCH

Tau

Timeframe	Single tau	
2025	HLT_tau160_mediumGNTau_L1eTAU140 Long-lived Particle ID - not scrutinized/validated in 2025 data HLT_tau180_mediumRNN_tracktwoLLP_L1eTAU140	HLT_tau25_mediumGNTau_tau20_mediumGN HLT_tau35_mediumGNTau_tau25_mediumGN HLT_tau35_mediumGNTau_tau25_mediumGN HLT_tau40_mediumGNTau_tau35_mediumGN HLT_tau80_mediumGNTau_tau35_mediumGN HLT_tau80_mediumGNTau_tau60_mediumGN Written to VBF Delayed Stream HLT_tau30_mediumGNTau_tau20_mediumGN HLT_tau30_mediumGNTau_tau20_mediumGN Long-lived Particle ID - not scrutinized/validated in 2025 data HLT_tau80_mediumRNN_tracktwoLLP_tau60_mediumGN

Electron/Muon + Tau

Trigger Group	Electron+Tau	
2025	HLT_e24_lhmedium_ivarloose_tau20_mediumGNTau_03dRAB_L1eEM26M HLT_e17_lhmedium_ivarloose_tau25_mediumGNTau_03dRAB_L1eEM18M_2cTAU20M_4jJ30	HLT_mu2 HLT_mu1

B-Jet + Tau

Trigger Group	B-Jet + Tau
2025	with b-jet and tau preselection; Main stream HLT_tau25_mediumGNTau_probe_L1eTAU12_j65c_020jvt_j40c_020jvt_j25c_020jvt_j20c_020jvt_SHARE HLT_tau25_mediumGNTau_probe_L1eTAU12_j65c_020jvt_j40c_020jvt_j25c_020jvt_j20c_020jvt_SHARE

MET (+ X)

Trigger Group	Pure MET	MET + X
From 400 bunches	HLT_xe65_cell_xe105_nn_L1jXE100 (recommended)	HLT_e70_lhloose_xe70_cell_L1eEM26M
	HLT_xe65_cell_xe90_pfopufit_L1jXE100	HLT_g90_loose_xe90_cell_L1eEM26M
	HLT_xe80_cell_xe115_tcpufit_L1jXE100	HLT_tau50_mediumGNTau_xe80_tcpufit_xe50_cell_L1jXE110
		HLT_tau50_mediumGNTau_xe80_pfopufit_xe50_cell_L1jXE110
		HLT_mu14_tau25_mediumGNTau_xe50_cell_03dRAB_L1MU
		HLT_tau60_mediumGNTau_tau25_mediumGNTau_xe50_cell_03dRAB_L1MU
		HLT_e17_lhmedium_tau25_mediumGNTau_xe50_cell_03dRAB_L1MU

Jets (DRAFT)

Trigger Group	Single jet	
75 bunches only	HLT_j175f_L1jJ90p30ETA49	N/A
75, 400 bunches	HLT_j420_L1jJ160 HLT_j420_ftf_preselj225_L1jJ160 Large-R HLT_j360_a10t_lcw_jes_L1jJ160 HLT_j360_a10t_lcw_jes_L1SC175-SCjJ10 HLT_j420_a10sd_cssk_pf_jes_ftf_preselj225_L1jJ160	HLT_2j250c_j120c_ftf_preselj225_L1jJ160 HLT_3j200_L1jJ160 HLT_3j200_ftf_presel3j160 HLT_4j115_ftf_presel4j80 HLT_5j70c_L14jJ40 HLT_5j70c_ftf_presel5j50 HLT_5j85_L14jJ40 HLT_5j85_ftf_presel5j50 HLT_6j55c_L14jJ40 HLT_6j55c_ftf_presel6j40 HLT_6j70_L14jJ40 HLT_6j70_ftf_presel6j40 HLT_7j45_L14jJ40 HLT_7j45_ftf_presel7j30 HLT_10j40_L14jJ40 HLT_10j40_ftf_presel7j30
From 400 bunches	Small-R HLT_j400_pf_ftf_preselj225_L1jJ160 HLT_j220f_L1jJ125p30ETA49 Large-R HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1jJ160 HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1SC175-SCjJ10 HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1gLJ140p0ETA25 HLT_j460_a10t_lcw_jes_L1jJ160 HLT_j460_a10t_lcw_jes_L1SC175-SCjJ10 HLT_j460_a10t_lcw_jes_L1gLJ140p0ETA25 HLT_j460_a10r_L1jJ160 HLT_j460_a10r_L1SC175-SCjJ10 HLT_j460_a10r_L1gLJ140p0ETA25 (**)	Small-R HLT_2j235c_j115c_pf_ftf_preselj225_L1jJ160 HLT_3j200_L1gJ400p0ETA25 HLT_3j200_pf_ftf_preselj225_L1jJ160 HLT_4j110_pf_ftf_preselj225_L1jJ160 HLT_5j65c_pf_ftf_preselj225_L1jJ160 HLT_5j70c_pf_ftf_preselj225_L1jJ160 HLT_5j80_pf_ftf_preselj225_L1jJ160 HLT_6j35c_020jvt_pf_ftf_preselj225_L1jJ160 HLT_6j55c_pf_ftf_preselj225_L1jJ160 HLT_6j65_pf_ftf_preselj225_L1jJ160

Trigger Group	Single jet	
	HLT_j460_a10_lcw_subjes_L1jJ160 HLT_j460_a10_lcw_subjes_L1SC175-SCjJ10 HLT_j460_a10_lcw_subjes_L1gLJ140p0ETA25 Large-R with mass cut HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1jJ160 HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1SC175-SCjJ10 HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1gLJ140p0ETA25 HLT_j420_35smcINF_a10t_lcw_jes_L1jJ160 HLT_j420_35smcINF_a10t_lcw_jes_L1SC175-SCjJ10 HLT_j420_35smcINF_a10t_lcw_jes_L1gLJ140p0ETA25 H_T HLT_j0_HT1000_L1HT190-jJ40s5pETA21 HLT_j0_HT1000_L1jJ160 HLT_j0_HT1000_L1gLJ400p0ETA25 HLT_j0_HT650XX0eta240_pf_ftf_preselcHT450_L1HT190-jJ40s5pETA21 HLT_j0_HT940_pf_ftf_preselcHT450_L1HT190-jJ40s5pETA21 HLT_j0_HT940_pf_ftf_preselj180_L1HT190-jJ40s5pETA21	HLT_7j45_pf_ftf_presel7 HLT_10j35_pf_ftf_prese Large-R with mass cut HLT_2j330_35smcINF_a HLT_2j330_35smcINF_a HLT_2j330_35smcINF_a HLT_2j330_35smcINF_a HLT_2j330_35smcINF_a HLT_2j330_35smcINF_a HLT_j360_60smcINF_j3 HLT_j360_60smcINF_j3 HLT_j370_35smcINF_j3 HLT_j370_35smcINF_j3

B-jets (DRAFT)

Trigger Group	Single b-jet
GN2	HLT_j210_0eta290_020jvt_bgn270_pf_ftf_preselj180_L1jJ160 HLT_j225_0eta290_020jvt_bgn277_pf_ftf_preselj180_L1jJ160 HLT_j275_0eta290_020jvt_bgn285_pf_ftf_preselj225_L1jJ160

Trigger Group	Multi b-jet
From 75 bunches	2b HLT_j165_0eta290_020jvt_bgn260_j55_0eta290_020jvt_bgn260_pf_ftf_preselj140bgtwo85XXj45bgtwo85 2b + 1j HLT_j140_2j50_0eta290_020jvt_bgn270_pf_ftf_preselj80XX2j45bgtwo90_L1jJ140_3jJ60 3b HLT_3j60_0eta290_020jvt_bgn277_pf_ftf_presel3j45bgtwo95_L13jJ70p0ETA23 1b + 3j HLT_j70_0eta290_020jvt_bgn260_3j70_pf_ftf_preselj50bgtwo85XX3j50_L14jJ50 2b + 2j, legacy HH HLT_2j35c_020jvt_bgn260_2j35c_020jvt_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25 2b + 2j

Trigger Group	Multi b-jet
	HLT_2j45_0eta290_020jvt_bgn260_2j45_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25
	2b + HT
	HLT_2j45_0eta290_020jvt_bgn270_j0_HT290_j0_DJMASS700j35_pf_ftf_L1HT150-jJ50s5pETA32_jMJ
	3b + 1j
	HLT_3j35_0eta290_020jvt_bgn270_j35_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25
	4b
	HLT_2j35_0eta290_020jvt_bgn270_2j35_0eta290_020jvt_bgn285_pf_ftf_presel4j25bgtwo95_L14jJ40p0ETA25
	HLT_4j35_0eta290_020jvt_bgn277_pf_ftf_presel4j25bgtwo95_L14jJ40p0ETA25
	2b + 3j
	HLT_2j35_0eta290_020jvt_bgn260_3j35_pf_ftf_presel3j25XX2j25bgtwo85_L15jJ40p0ETA25
	1b+5j
	HLT_5j35c_020jvt_j25c_020jvt_SHARED_j25c_020jvt_bgn260_pf_ftf_presel5c25XXc25bgtwo85_L14jJ40p0ETA25

B-physics (DRAFT)

Trigger Group	K-star ee Triggers
From 75, 400 bunches	HLT_e5_lhvloose_bBeeM6000_L1BKeePrimary HLT_2e5_lhvloose_bBeeM6000_L1BKeePrimary HLT_e5_lhvloose_bBeeM6000_L1BKeePrescaled HLT_2e5_lhvloose_bBeeM6000_L1BKeePrescaled

Trigger Group	B-physics muon triggers	B-physics muon triggers
From 75 bunches	Dimuon HLT_2mu10_bJpsimumu_L12MU8F HLT_2mu10_bUpsimumu_L12MU8F HLT_mu11_mu6_bBmumu_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuD0Xloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDp0loose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuPi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BdmumuKst_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BpmumuKp_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_LbPqKm_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu11_mu6_bPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bTau_L1MU8VF_2MU5VF	Dimuon HLT_2mu6_bBmumu_L12MU5VF HLT_2mu6_bDimu_L12MU5VF HLT_2mu6_bPhi_L12MU5VF HLT_2mu4_bBmumux_BcmumuD0Xloose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuDp0loose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuPi_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BdmumuKst_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BpmumuKp_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_LbPqKm_L1MU8VF_2MU5VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bDimu2700_L1MU8VF_2MU5VF HLT_2mu4_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bJpsimumu_L12MU3VF HLT_2mu4_bJpsimumu_Lxy0_L12MU3VF HLT_2mu4_bJpsimumu_sigmaLxy3_L12MU3VF HLT_2mu4_bUpsimumu_L12MU3VF HLT_2mu4_bBmumu_L12MU3VF HLT_2mu4_bBmumu_Lxy0_L12MU3VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bDimu2700_sigmaLxy3_L12MU3VF

Trigger Group	B-physics muon triggers	B-physics muon triggers
	HLT_mu11_mu6_bUpsimumu_L1MU8VF_2MU5VF HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF Support HLT_mu11_l2io_mu6_l2io_invmDimu_L1MU8VF_2MU5VF Dimuon L1Topo HLT_2mu6_bJpsimumu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bJpsimumu_Lxy0_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bBmumu_Lxy0_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bUpsimumu_L1BPH-8M15-0DR22-2MU5VF HLT_mu11_mu6_bJpsimumu_L1LFV-MU8VF HLT_mu11_mu6_bJpsimumu_sigmaLxy3_L1LFV-MU8VF HLT_mu11_mu6_bUpsimumu_L1LFV-MU8VF HLT_mu11_mu6_bBmumu_L1LFV-MU8VF HLT_mu11_mu6_bDimu_L1LFV-MU8VF HLT_mu11_mu6_bDimu2700_L1LFV-MU8VF HLT_mu11_mu6_bDimu2700_sigmaLxy3_L1LFV-MU8VF HLT_mu11_mu6_bPhi_L1LFV-MU8VF HLT_mu11_mu6_bTau_L1LFV-MU8VF Support HLT_mu11_l2io_mu6_l2io_invmDimu_L1LFV-MU8VF 3 muons HLT_3mu4_b3mu_noos_L13MU3VF HLT_3mu4_bJpsi_L13MU3VF HLT_3mu4_bPhi_L13MU3VF HLT_3mu4_bTau_L13MU3VF HLT_3mu4_bUpsi_L13MU3VF HLT_3mu6_bJpsi_L13MU5VF HLT_3mu6_bUpsi_L13MU5VF HLT_3mu6_bDimu_L13MU5VF HLT_3mu6_bTau_L13MU5VF HLT_2mu6_mu4_bTau_L12MU5VF_3MU3V HLT_2mu6_mu4_bUpsi_L12MU5VF_3MU3V HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF 4 muons HLT_4mu4_bDimu6000_L14MU3V Multiple-candidate L1 flag HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU10BOM HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU12BOM	HLT_mu6_mu4_bJpsimumu_L1MU8VF_2MU5VF HLT_mu6_mu4_bJpsimumu_Lxy0_L1MU8VF_2MU5VF HLT_mu6_mu4_bJpsimumu_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu6_mu4_bBmumu_L1MU8VF_2MU5VF HLT_mu6_mu4_bBmumu_Lxy0_L1MU8VF_2MU5VF HLT_mu6_mu4_bDimu2700_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu6_mu4_bUpsimumu_L1MU8VF_2MU5VF Support HLT_2mu6_bBmumu_Lxy0_L12MU5VF_3MU3V HLT_2mu6_bJpsimumu_Lxy0_L12MU5VF_3MU3V Dimuon L1Topo HLT_2mu6_bBmumu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bDimu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bJpsimumu_sigmaLxy3_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bDimu_L1LFV-MU8VF HLT_2mu6_bDimu2700_sigmaLxy3_L1LFV-MU8VF HLT_2mu6_bPhi_L1LFV-MU8VF HLT_mu6_mu4_bBmumu_Lxy0_L1LFV-MU8VF HLT_mu6_mu4_bUpsimumu_L1LFV-MU8VF

Trigger Group	B-physics muon triggers	B-physics muon triggers
	Bmux HLT_mu20_bBmux_BdmuDpX_L1MU14FCH HLT_mu20_bBmux_BdmuDstarX_L1MU14FCH HLT_mu20_bBmux_BpmuD0X_L1MU14FCH HLT_mu20_bBmux_BsmuDsX_L1MU14FCH HLT_mu20_bBmux_LbmuLcX_L1MU14FCH	

VBF (DRAFT)

Trigger Group	B-jet + X
From 75 bunches	B-jet + forward jet HLT_j50_0eta290_020jvt_bgn270_2j45f_pf_ftf_preselj45XX2f40_L1jJ55p0ETA23_2jJ40p30ETA49 Photon + jet HLT_g25_medium_4j35a_j0_DJMASS1000j35_L1eEM26M Photon + b-jet HLT_g20_tight_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS500j35_pf_ftf_L1eEM22M_jMJJ-300 (sin HLT_g25_medium_2j35_0eta290_020jvt_bgn277_2j35a_pf_ftf_presel2a20bgtwo90XX2a20_L1eEM26M HLT_g25_medium_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS700j35_pf_ftf_presela20bgtwo85XX3a20 HLT_g25_tight_2j35_0eta290_020jvt_bgn277_2j35a_pf_ftf_presel2a20bgtwo90XX2a20_L1eEM26M HLT_g25_tight_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS700j35_pf_ftf_presela20bgtwo85XX3a20

Trigger Group	Dijet mass L1
From 75 bunches	Main stream HLT_e10_lhmedium_ivarloose_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_mu10_ivarmedium_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_tau25_mediumRNN_tracktwoMVA_tau20_mediumRNN_tracktwoMVA_03dRAB_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF Delayed stream HLT_e5_lhvloose_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1jMJJ-500-NFF HLT_mu4_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1jMJJ-500-NFF HLT_j70_j50a_j0_DJMASS1000j50dphi240_xe90_tcpufit_xe50_cell_L1jMJJ-500-NFF HLT_j70_j50a_j0_DJMASS1000j50dphi200x400deta_L1jMJJ-500-NFF HLT_2mu6_2j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_2e5_lhmedium_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_j65a_j45a_2j35a_SHARED_2j35_0eta290_020jvt_bgn270_j0_DJMASS1000j50_pf_ftf_presela60XX2a20

Unconventional tracking (LLP) (DRAFT)

Trigger Group	Unconventional tracking
From 75 bunches	Hit-based displaced vertex HLT_xe80_tcpufit_hitdvjet200_tight_L1jXE100 HLT_j180_hitdvjet260_tight_L1jJ160 Prescaled until period H1, then unprescaled

Trigger Group	Unconventional tracking
	High dE/dX HLT_xe80_tcpufit_dedxtrk50_medium_L1jXE100 Disappearing track HLT_xe80_tcpufit_distrk20_medium_L1jXE100 HLT_xe80_tcpufit_distrk20_tight_L1jXE100 Isolated track HLT_xe80_tcpufit_isotrk120_medium_iaggrmedium_L1jXE100 HLT_xe80_tcpufit_isotrk140_medium_iaggrmedium_L1jXE100 Displaced jet HLT_j180_2dispjet50_3d2p_L1jJ160 HLT_j180_dispjet50_3d2p_dispjet50_1p_L1jJ160 HLT_j180_dispjet90_x3d1p_L1jJ160 HLT_j180_dispjet100_x3d1p_L1jJ160

Dedicated physics triggers (DRAFT)

Trigger Group	Standard Model
75 bunches only	Low-mass Drell-Yan HLT_mu4_ivarloose_mu4_11invAB60_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_7invAB9_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b11invAB60vtx20_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b7invAB9vtx20_L1DY-BOX-2MU3VF
75, 400 bunches	HLT_mu6_ivarloose_mu6_11invAB24_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_24invAB60_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b11invAB24vtx20_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b24invAB60vtx20_L1DY-BOX-2MU5VF

Trigger Group	Higgs
From 75 bunches	Merged electron HLT_e25_mergedtight_g35_medium_90invAB_02dRAB_L1eEM24L Photon+meson HLT_g35_medium_tau25_dipion3_tracktwoMVA_L1eTAU12_60invAB_L1eEM26M HLT_g25_medium_tau25_singlepion_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dipion1_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dipion2_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dikaonmass_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_kaonpi1_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_kaonpi2_tracktwoMVA_50invAB_L1eEM26M

Trigger Group	HIGP
From 75 bunches	HH 4b jet-seeded 3b1j, Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_3j20c_020jvt_bgn282_pf_ftf_presel2c jet-seeded 2b2j

Trigger Group	HIGP
	Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn277_pf_ftf_presel2c VBFDelayed stream before Run 474770, Main stream after Run 474770 HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn280_pf_ftf_presel2c muon-seeded 2b2j Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn277_pf_ftf_presel2c VBFDelayed stream before Run 474770, Main stream after Run 474770 HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn280_pf_ftf_presel2c

Trigger Group	Exotics
From 75 bunches	Non-colliding bunches HLT_2g22_tight_L1eEM15_EMPTY HLT_2g22_tight_L1eEM9_UNPAIRED_ISO HLT_2g50_tight_L1eEM15_EMPTY HLT_2g50_tight_L1eEM9_UNPAIRED_ISO HLT_g35_medium_g25_medium_L1eEM15_EMPTY HLT_g35_medium_g25_medium_L1eEM9_UNPAIRED_ISO
From 75 bunches	Dijet + ISR HLT_g85_tight_3j50_L1eEM26M Mono-pole HLT_g0_hiptrt_L1eEM26M LLP, colliding bunches HLT_2mu50_msonly_L1MU14FCH HLT_g40_loose_L1eEM24L_mu40_msonly_L1MU14FCH HLT_g15_loose_L1eEM10L_2mu10_msonly_L12MU8F HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_jJ90 HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_jXE80 Non-colliding bunches HLT_g15_loose_L1eEM10L_2mu10_msonly_L1MU3V_EMPTY HLT_g15_loose_L1eEM10L_2mu10_msonly_L1MU3V_UNPAIRED_ISO HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF_EMPTY HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF_UNPAIRED_ISO HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_EMPTY HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_UNPAIRED_ISO

Trigger Group	Supersymmetry
From 75 bunches	B-jet + MET (sbottom) HLT_j75_0eta290_020jvt_bgn260_pf_ftf_xe60_cell_-L12jJ90_jXE80 HLT_j95_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_pfopufit_L1jXE110 HLT_j95_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_tcpufit_L1jXE110 HLT_2j45_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_pfopufit_L12jJ40_jXE110 HLT_2j45_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_tcpufit_L12jJ40_jXE110

Trigger Group	Supersymmetry
	HLT_3j35_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe70_pfpufit_L13jJ40p0ETA25_jXE80 HLT_3j35_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe70_tcpufit_L13jJ40p0ETA25_jXE80
	Photon + multijet (stealth SUSY) HLT_g45_loose_6j45c_L14jJ40p0ETA25

Triggers in 2024

Introduction

Recommended triggers for 2024 physics analyses can be found below. Known issues are detailed in [TriggerIssues2024](#).

See [TriggerNamingRun3](#) for naming conventions.

Trigger periods in intensity ramp-up (DRAFT)

Rulebook branches listed are the final tag used for that period, which contain the most stringent rules applied.

Intensity step	Data periods	First run number	Rulebook branch
2 bunches	C1	470781	RB-r24p0p28 ↗
8 bunches	C2	472056	RB-r24p0p28 ↗
4 bunches	D1,D2	472553	RB-r24p0p32 ↗
12 bunches	D3,D4	472604	RB-r24p0p32 ↗
75b bunches	D5,D6,D7	472662	RB-r24p0p32 ↗
400 bunches	D8,D9,D10	472943:472968	RB-r24p0p32 ↗
4 bunches	J1	478573	RB-r24p0p47 ↗
75 bunches	J2,J3	478583	RB-r24p0p47 ↗
400 bunches	J4	478607:478609	RB-r24p0p47 ↗
1200 bunches	J5	478634	RB-r24p0p47b ↗
2400 bunches	J6,J7	478653:478663	RB-r24p0p47 ↗
2400 bunches	J8	478763	RB-r24p0p47-magnetsOff ↗
400 bunches	L1	480146: 480161	RB-r24p0p53 ↗
2400 bunches	P1,P2,P3	486794	RB-r24p0p65 ↗
8:62 bunches	Q1	488210:488239	RB-r24p0p66p1 ↗
400 bunches	Q2,Q3	488393:488427	RB-r24p0p66p1 ↗
2400 bunches	Q4	488474:488589	RB-r24p0p66p1 ↗
2400 bunches	O1:O17	484909	RB-r24p0p62 ↗
1800 bunches	O18	486384	RB-r24p0p62 ↗
bunches	O20,O21,O22	486640	RB-r24p0p62 ↗
2400 bunches	N1:N38	482374	RB-r24p0p56 ↗
2400 bunches	M1:M33	480188:480277	RB-r24p0p53 ↗
2400 bunches	K1:K18	479103	RB-r24p0p49 ↗

For runs in each data periods, see [COMA](#) [↗](#)

Egamma

Trigger Group	Single electron	Multi electron
From 75 bunches	HLT_e26_lhtight_ivarloose_L1eEM26M HLT_e26_lhtight_ivarloose_L1eEM26T HLT_e20_lhtight_ivarloose_L1ZAFB-25DPHI-eEM18M HLT_e26_lhtight_ivarmedium_L1eEM26M HLT_e26_lhtight_ivartight_L1eEM26M HLT_e60_lhmedium_L1eEM26M HLT_e140_lhloose_L1eEM26M HLT_e140_lhloose_noringer_L1eEM26M HLT_e300_etcut_L1eEM26M Large-radius tracking HLT_e30_lhloose_nopix_lrtmedium_L1eEM26M	HLT_2e17_lhvloose_L12eEM18M HLT_2e24_lhvloose_L12eEM24L HLT_e24_lhvloose_2e12_lhvloose_L1eEM24L_3 HLT_e26_lhtight_e14_idperf_tight_probe_50inv HLT_e26_lhtight_e14_idperf_tight_nogsf_probe_50 HLT_e26_lhtight_e14_etcut_probe_50invmAB13

Trigger Group	Single photon	Multi photon
From 75 bunches	HLT_g140_loose_L1eEM26M HLT_g300_etcut_L1eEM26M	HLT_2g22_tight_L12eEM18M HLT_2g50_loose_L12eEM24L HLT_2g20_tight_icaloose_L12eEM18M HLT_g35_medium_g25_medium_L12eEM24L HLT_2g25_loose_g15_loose_L12eEM24L HLT_2g15_tight_25dphiAA_invAA80_L1DPHI-M70-2eEM15M
From Run 476718		HLT_g45_medium_g20_medium_L1eEM40L_2eEM18L

Trigger Group	Electron + photon
From 75 bunches	HLT_e24_lhmedium_g25_medium_02dRAB_L12eEM24L HLT_e24_lhmedium_g12_loose_g12_loose_02dRAB_02dRAC_02dRBC_L1eEM24L_3eEM12L HLT_e25_mergedtight_g35_medium_90invmAB_02dRAB_L12eEM24L

Muon

Trigger Group	Single muon	Multi muon
75, 400 bunches only	N/A	N/A
From 900 bunches	HLT_mu24_ivarmedium_L1MU14FCH HLT_mu50_L1MU14FCH HLT_mu60_0eta105_msonly_L1MU14FCH HLT_mu60_L1MU14FCH HLT_mu80_msonly_3layersEC_L1MU14FCH HLT_mu3vtx_L12MU8F	HLT_2mu14_L12MU8F HLT_mu20_ivarmedium_mu8noL1_L1MU14FCH HLT_mu22_mu8noL1_L1MU14FCH HLT_3mu6_L13MU5VF HLT_4mu4_L14MU3V HLT_2mu10_l2mt_L1MU10BOM

Trigger Group	Single muon	Multi muon
	Large-radius tracking HLT_mu20_LRT_d0loose_L1MU14FCH	Invariant mass cut HLT_mu10_ivarmedium_mu10_10invmAB70_L12MU8F HLT_mu20_ivarmedium_mu4noL1_10invmAB70_L1MU14FCH

Electron/Photon + Muon

Trigger Group	Electron + Muon	Photon + Muon
From 400 bunches	N/A	Photon + 2 muon HLT_g35_loose_mu15_mu2noL1_L1eEM28M
From 900 bunches	Electron + muon HLT_e7_lhmedium_L1eEM5_mu24_L1MU14FCH HLT_e17_lhloose_mu14_L1eEM18L_MU8F HLT_e26_lhmedium_mu8noL1_L1eEM26M 2 electron + muon HLT_2e12_lhloose_mu10_L12eEM10L_MU8F Electron + 2muon HLT_e9_lhvloose_L1eEM5_mu20_mu8noL1_L1MU14FCH HLT_e12_lhloose_L1eEM10L_2mu10_L12MU8F	Photon + muon HLT_g25_medium_L1eEM18L_mu24_L1MU14FCH HLT_g25_medium_L1eEM26M_mu24_ivarmedium_mu10_10invmAB70_L12MU8F HLT_g35_loose_mu18_L1eEM28M HLT_g35_loose_mu15_mu2noL1_L1eEM28M HLT_g40_loose_L1eEM24L_mu40_msonly_mu10_L12MU8F 2 photon + muon HLT_2g10_loose_L1eEM9_mu20_L1MU14FCH

Tau

Timeframes are defined in this wikpage: [TauTriggerChanges2024](#)

Timeframe	Single tau	
2024	HLT_tau160_mediumRNN_tracktwoMVA_L1eTAU140 Long-lived Particle ID - not scrutinized/validated in 2024 data HLT_tau180_mediumRNN_tracktwoLLP_L1eTAU140	HLT_tau25_mediumRNN_tracktwoMVA_tau25_L1eTAU140 HLT_tau35_mediumRNN_tracktwoMVA_tau35_L1eTAU140 HLT_tau35_mediumRNN_tracktwoMVA_tau35_L1eTAU140 HLT_tau40_mediumRNN_tracktwoMVA_tau40_L1eTAU140 HLT_tau80_mediumRNN_tracktwoMVA_tau80_L1eTAU140 HLT_tau80_mediumRNN_tracktwoMVA_tau80_L1eTAU140 Written to VBF Delayed Stream HLT_tau30_mediumRNN_tracktwoMVA_tau30_L1eTAU140 HLT_tau30_mediumRNN_tracktwoMVA_tau30_L1eTAU140 Long-lived Particle ID - not scrutinized/validated in 2024 data HLT_tau80_mediumRNN_tracktwoLLP_tau80_L1eTAU140

Trigger Group	Electron+Tau
2024	HLT_e24_lhmedium_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1eEM26M
Timeframes A to C	HLT_e17_lhmedium_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1eEM18M_2eTAU20M
Timeframes D to F	HLT_e17_lhmedium_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1eEM18M_2cTAU20M

MET (+ X)

Trigger Group	Pure MET	
From 400 bunches	HLT_xe65_cell_xe105_nn_L1jXE100 (recommended) HLT_xe65_cell_xe90_pfopufit_L1jXE100 HLT_xe80_cell_xe115_tcpufit_L1jXE100	HLT_e70_lhloose_xe70_cell_L1eEM26M HLT_g90_loose_xe90_cell_L1eEM26M HLT_tau50_mediumRNN_tracktwoMVA_xe80_tcpufit_xe50 HLT_tau50_mediumRNN_tracktwoMVA_xe80_pfopufit_xe50
Timeframes A to C		HLT_tau60_mediumRNN_tracktwoMVA_tau25_mediumRNN HLT_e17_lhmedium_tau25_mediumRNN_tracktwoMVA_xe50
Timeframes B to F		HLT_mu14_tau25_mediumRNN_tracktwoMVA_xe50_cell_
Timeframes D to F		HLT_tau60_mediumRNN_tracktwoMVA_tau25_mediumRNN HLT_e17_lhmedium_tau25_mediumRNN_tracktwoMVA_xe50

Note that all triggers listed here were run unprescaled in timeframes specified. These timeframes are defined in TauTriggerChanges2024. The definition of cTAU seeds changed also according to timeframes listed in this [wikipage](#).

Jets

Trigger Group	Single jet	
75 bunches only	HLT_j175f_L1jJ90p30ETA49	N/A
75, 400 bunches	HLT_j420_L1jJ160 HLT_j420_ftf_preselj225_L1jJ160 Large-R HLT_j360_a10t_lcw_jes_L1jJ160 HLT_j360_a10t_lcw_jes_L1SC111-CjJ40 HLT_j420_a10sd_cssk_pf_jes_ftf_preselj225_L1jJ160	HLT_2j250c_j120c_ftf_presel HLT_3j200_L1jJ160 HLT_3j200_ftf_presel3j150_L1 HLT_4j115_ftf_presel4j85_L1 HLT_5j70c_L14jJ40 HLT_5j70c_ftf_presel5j50_L1 HLT_5j85_L14jJ40 HLT_5j85_ftf_presel5j50_L14 HLT_6j55c_L14jJ40 HLT_6j55c_ftf_presel6j40_L1 HLT_6j70_L14jJ40 HLT_6j70_ftf_presel6j40_L14 HLT_7j45_L14jJ40 HLT_7j45_ftf_presel7j30_L14 HLT_10j40_L14jJ40 HLT_10j40_ftf_presel7j30_L1
From 400 bunches	Small-R HLT_j400_pf_ftf_preselj225_L1jJ160	Small-R HLT_2j235c_j115c_pf_ftf_pre HLT_3j200_L1gJ400p0ETA2

Trigger Group	Single jet	
	HLT_j220f_L1j125p30ETA49 Large-R HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1j160 HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CjJ40 (*) HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1gLJ140p0ETA25 (**) HLT_j460_a10t_lcw_jes_L1j160 HLT_j460_a10t_lcw_jes_L1SC111-CjJ40 (*) HLT_j460_a10t_lcw_jes_L1gLJ140p0ETA25 (**) HLT_j460_a10r_L1j160 HLT_j460_a10r_L1SC111-CjJ40 (*) HLT_j460_a10r_L1gLJ140p0ETA25 (**) HLT_j460_a10_lcw_subjes_L1j160 HLT_j460_a10_lcw_subjes_L1SC111-CjJ40 (*) HLT_j460_a10_lcw_subjes_L1gLJ140p0ETA25 (**) Large-R with mass cut HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1j160 HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CjJ40 (*) HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1gLJ140p0ETA25 (**) HLT_j420_35smcINF_a10t_lcw_jes_L1j160 HLT_j420_35smcINF_a10t_lcw_jes_L1SC111-CjJ40 (*) HLT_j420_35smcINF_a10t_lcw_jes_L1gLJ140p0ETA25 (**) H_T HLT_j0_HT1000_L1HT190-jJ40s5pETA21 HLT_j0_HT1000_L1j160 HLT_j0_HT1000_L1gJ400p0ETA25 HLT_j0_HT650XX0eta240_pf_ftf_preselcHT450_L1HT190-jJ40s5pETA21 HLT_j0_HT940_pf_ftf_preselcHT450_L1HT190-jJ40s5pETA21 HLT_j0_HT940_pf_ftf_preselj180_L1HT190-jJ40s5pETA21	HLT_3j200_pf_ftf_presel3j15 HLT_4j110_pf_ftf_presel4j85 HLT_5j65c_pf_ftf_presel5c50 HLT_5j70c_pf_ftf_presel5j50 HLT_5j80_pf_ftf_presel5j50 HLT_6j35c_020jvt_pf_ftf_pre HLT_6j55c_pf_ftf_presel6j40 HLT_6j65_pf_ftf_presel6j40 HLT_7j45_pf_ftf_presel7j30 HLT_10j35_pf_ftf_presel7j30 Large-R with mass cut HLT_2j330_35smcINF_a10sd HLT_2j330_35smcINF_a10sd HLT_2j330_35smcINF_a10sd HLT_2j330_35smcINF_a10t HLT_2j330_35smcINF_a10t HLT_2j330_35smcINF_a10t HLT_j360_60smcINF_j360_a HLT_j360_60smcINF_j360_a HLT_j370_35smcINF_j370_a HLT_j370_35smcINF_j370_a

* The large-R jet L1 item SC111-CjJ40 were prescaled in a few runs due to high rate. See the StableBeamRuns2024 twiki for more details.

** The large-R jet L1 item gLJ were disabled in a few runs due to low IBL limit. See the StableBeamRuns2024 twiki for more details

B-jets

Trigger Group	Single b-jet
GN2	HLT_j210_0eta290_020jvt_bgn270_pf_ftf_preselj180_L1j160 HLT_j225_0eta290_020jvt_bgn277_pf_ftf_preselj180_L1j160 HLT_j275_0eta290_020jvt_bgn285_pf_ftf_preselj225_L1j160

Trigger Group	Multi b-jet
From 75 bunches	2b HLT_j165_0eta290_020jvt_bgn260_j55_0eta290_020jvt_bgn260_pf_ftf_preselj140bgtwo85XXj45bgtwo85
	2b + 1j HLT_j140_2j50_0eta290_020jvt_bgn270_pf_ftf_preselj80XX2j45bgtwo90_L1j140_3jJ60
	3b HLT_3j60_0eta290_020jvt_bgn277_pf_ftf_presel3j45bgtwo95_L13jJ70p0ETA23
	1b + 3j HLT_j70_0eta290_020jvt_bgn260_3j70_pf_ftf_preselj50bgtwo85XX3j50_L14jJ50
	2b + 2j, legacy HH HLT_2j35c_020jvt_bgn260_2j35c_020jvt_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25
	2b + 2j HLT_2j45_0eta290_020jvt_bgn260_2j45_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25
	2b + HT HLT_2j45_0eta290_020jvt_bgn270_j0_HT290_j0_DJMASS700j35_pf_ftf_L1HT150-jJ50s5pETA32_jMJ
	3b + 1j HLT_3j35_0eta290_020jvt_bgn270_j35_pf_ftf_presel2j25XX2j25bgtwo85_L14jJ40p0ETA25
	4b HLT_2j35_0eta290_020jvt_bgn270_2j35_0eta290_020jvt_bgn285_pf_ftf_presel4j25bgtwo95_L14jJ40p0ETA25 HLT_4j35_0eta290_020jvt_bgn277_pf_ftf_presel4j25bgtwo95_L14jJ40p0ETA25
	2b + 3j HLT_2j35_0eta290_020jvt_bgn260_3j35_pf_ftf_presel3j25XX2j25bgtwo85_L15jJ40p0ETA25
	1b+5j HLT_5j35c_020jvt_j25c_020jvt_SHARED_j25c_020jvt_bgn260_pf_ftf_presel5c25XXc25bgtwo85_L14jJ4

B-physics

Trigger Group	K-star ee Triggers
From 75, 400 bunches	HLT_e5_lhvloose_bBeeM6000_L1BKeePrimary
	HLT_2e5_lhvloose_bBeeM6000_L1BKeePrimary
	HLT_e5_lhvloose_bBeeM6000_L1BKeePrescaled
	HLT_2e5_lhvloose_bBeeM6000_L1BKeePrescaled

Trigger Group	B-physics muon triggers	B-physics muon triggers
From 75 bunches	Dimuon	Dimuon
	HLT_2mu10_bJpsimumu_L12MU8F	HLT_2mu6_bBmumu_L12MU5V
	HLT_2mu10_bUpsimumu_L12MU8F	HLT_2mu6_bDimu_L12MU5VF
	HLT_mu11_mu6_bBmumu_L1MU8VF_2MU5VF	HLT_2mu6_bPhi_L12MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuD0Xloose_L1MU8VF_2MU5VF	HLT_2mu4_bBmumux_Bcmumu
	HLT_mu11_mu6_bBmumux_BcmumuDp0loose_L1MU8VF_2MU5VF	HLT_2mu4_bBmumux_Bcmumu

Trigger Group	B-physics muon triggers	B-physics muon triggers
	HLT_mu11_mu6_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuPi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BdmumuKst_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BpmumuKp_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_LbPqKm_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu11_mu6_bPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bTau_L1MU8VF_2MU5VF HLT_mu11_mu6_bUpsimumu_L1MU8VF_2MU5VF HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_sigmaLxy3_L1MU8VF_2MU5VF HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF Support HLT_mu11_l2io_mu6_l2io_invmDimu_L1MU8VF_2MU5VF Dimuon L1Topo HLT_2mu6_bJpsimumu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bJpsimumu_Lxy0_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bBmumu_Lxy0_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bUpsimumu_L1BPH-8M15-0DR22-2MU5VF HLT_mu11_mu6_bJpsimumu_L1LFV-MU8VF HLT_mu11_mu6_bJpsimumu_sigmaLxy3_L1LFV-MU8VF HLT_mu11_mu6_bUpsimumu_L1LFV-MU8VF HLT_mu11_mu6_bBmumu_L1LFV-MU8VF HLT_mu11_mu6_bDimu_L1LFV-MU8VF HLT_mu11_mu6_bDimu2700_L1LFV-MU8VF HLT_mu11_mu6_bDimu2700_sigmaLxy3_L1LFV-MU8VF HLT_mu11_mu6_bPhi_L1LFV-MU8VF HLT_mu11_mu6_bTau_L1LFV-MU8VF Support HLT_mu11_l2io_mu6_l2io_invmDimu_L1LFV-MU8VF 3 muons HLT_3mu4_b3mu_noos_L13MU3VF HLT_3mu4_bJpsi_L13MU3VF HLT_3mu4_bPhi_L13MU3VF HLT_3mu4_bTau_L13MU3VF HLT_3mu4_bUpsi_L13MU3VF HLT_3mu6_bJpsi_L13MU5VF HLT_3mu6_bUpsi_L13MU5VF	HLT_2mu4_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BcmumuPi_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BdmumuKst_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BpmumuKp_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF HLT_2mu4_bBmumux_LbPqKm_L1MU8VF_2MU5VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bJpsimumu_L12MU3VF HLT_2mu4_bJpsimumu_Lxy0_L12MU3VF HLT_2mu4_bJpsimumu_sigmaLxy3_L12MU3VF HLT_2mu4_bUpsimumu_L12MU3VF HLT_2mu4_bBmumu_L12MU3VF HLT_2mu4_bBmumu_Lxy0_L12MU3VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bDimu2700_sigmaLxy3_L12MU3VF HLT_mu6_mu4_bJpsimumu_L12MU3VF HLT_mu6_mu4_bJpsimumu_Lxy0_L12MU3VF HLT_mu6_mu4_bJpsimumu_sigmaLxy3_L12MU3VF HLT_mu6_mu4_bBmumu_L12MU3VF HLT_mu6_mu4_bBmumu_Lxy0_L12MU3VF HLT_mu6_mu4_bDimu2700_sigmaLxy3_L12MU3VF HLT_mu6_mu4_bUpsimumu_L12MU3VF Support HLT_2mu6_bBmumu_Lxy0_L12MU3VF HLT_2mu6_bJpsimumu_Lxy0_L12MU3VF Dimuon L1Topo HLT_2mu6_bBmumu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bDimu_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bJpsimumu_sigmaLxy3_L1BPH-2M9-2DR15-2MU5VF HLT_2mu6_bDimu_L1LFV-MU8VF HLT_2mu6_bDimu2700_sigmaLxy3_L1LFV-MU8VF HLT_2mu6_bPhi_L1LFV-MU8VF HLT_mu6_mu4_bBmumu_Lxy0_L1LFV-MU8VF HLT_mu6_mu4_bUpsimumu_L1LFV-MU8VF

Trigger Group	B-physics muon triggers	B-physics muon triggers
	HLT_3mu6_bDimu_L13MU5VF HLT_3mu6_bTau_L13MU5VF HLT_2mu6_mu4_bTau_L12MU5VF_3MU3V HLT_2mu6_mu4_bUpsi_L12MU5VF_3MU3V HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF 4 muons HLT_4mu4_bDimu6000_L14MU3V Multiple-candidate L1 flag HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU10BOM HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU12BOM Bmux HLT_mu20_bBmux_BdmuDpX_L1MU14FCH HLT_mu20_bBmux_BdmuDstarX_L1MU14FCH HLT_mu20_bBmux_BpmuD0X_L1MU14FCH HLT_mu20_bBmux_BsmuDsX_L1MU14FCH HLT_mu20_bBmux_LbmuLcX_L1MU14FCH	

VBF

Trigger Group	B-jet + X
From 75 bunches	B-jet + forward jet HLT_j50_0eta290_020jvt_bgn270_2j45f_pf_ftf_preselj45XX2f40_L1jJ55p0ETA23_2jJ40p30ETA49 Photon + jet HLT_g25_medium_4j35a_j0_DJMASS1000j35_L1eEM26M Photon + b-jet HLT_g20_tight_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS500j35_pf_ftf_L1eEM22M_jMJJ-300 (sin HLT_g25_medium_2j35_0eta290_020jvt_bgn277_2j35a_pf_ftf_presel2a20bgtwo90XX2a20_L1eEM26M HLT_g25_medium_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS700j35_pf_ftf_presela20bgtwo85XX3a20 HLT_g25_tight_2j35_0eta290_020jvt_bgn277_2j35a_pf_ftf_presel2a20bgtwo90XX2a20_L1eEM26M HLT_g25_tight_j35_0eta290_020jvt_bgn277_3j35a_j0_DJMASS700j35_pf_ftf_presela20bgtwo85XX3a20

Trigger Group	Dijet mass L1
From 75 bunches	Main stream HLT_e10_lhmedium_ivarloose_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_mu10_ivarmedium_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_tau25_mediumRNN_tracktwoMVA_tau20_mediumRNN_tracktwoMVA_03dRAB_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF Delayed stream HLT_e5_lhvloose_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1jMJJ-500-NFF

Trigger Group	Dijet mass L1
	HLT_mu4_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1jMJJ-500-NFF HLT_j70_j50a_j0_DJMASS1000j50dphi240_xe90_tcpufit_xe50_cell_L1jMJJ-500-NFF HLT_j70_j50a_j0_DJMASS1000j50dphi200x400deta_L1jMJJ-500-NFF HLT_2mu6_2j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_2e5_1hmedium_j70_j50a_j0_DJMASS900j50_L1jMJJ-500-NFF HLT_j65a_j45a_2j35a_SHARED_2j35_0eta290_020jvt_bgn270_j0_DJMASS1000j50_pf_ftf_presela60XX

Unconventional tracking (LLP)

Trigger Group	Unconventional tracking
From 75 bunches	Hit-based displaced vertex HLT_xe80_tcpufit_hitdvjet200_tight_L1jXE100 HLT_j180_hitdvjet260_tight_L1jJ160 Prescaled until period H1, then unprescaled High dE/dX HLT_xe80_tcpufit_dedxtrk50_medium_L1jXE100 Disappearing track HLT_xe80_tcpufit_distrk20_medium_L1jXE100 HLT_xe80_tcpufit_distrk20_tight_L1jXE100 Isolated track HLT_xe80_tcpufit_isotrk120_medium_iaggrmedium_L1jXE100 HLT_xe80_tcpufit_isotrk140_medium_iaggrmedium_L1jXE100 Displaced jet HLT_j180_2dispjet50_3d2p_L1jJ160 HLT_j180_dispjet50_3d2p_dispjet50_1p_L1jJ160 HLT_j180_dispjet90_x3d1p_L1jJ160 HLT_j180_dispjet100_x3d1p_L1jJ160

Dedicated physics triggers

Trigger Group	Standard Model
75 bunches only	Low-mass Drell-Yan HLT_mu4_ivarloose_mu4_11invAB60_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_7invAB9_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b11invAB60vtx20_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b7invAB9vtx20_L1DY-BOX-2MU3VF
75, 400 bunches	HLT_mu6_ivarloose_mu6_11invAB24_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_24invAB60_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b11invAB24vtx20_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b24invAB60vtx20_L1DY-BOX-2MU5VF

Trigger Group	Higgs
From 75 bunches	Merged electron HLT_e25_mergedtight_g35_medium_90invAB_02dRAB_L12eEM24L

Trigger Group	Higgs
	Photon+meson HLT_g35_medium_tau25_dipion3_tracktwoMVA_60invmAB_L1eEM26M (before Run 480957) HLT_g35_medium_tau25_dipion3_tracktwoMVA_L1eTAU12_60invmAB_L1eEM26M (after Run 480957) HLT_g25_medium_tau25_singlepion_tracktwoMVA_50invmAB_L1eEM26M HLT_g25_medium_tau25_dipion1_tracktwoMVA_50invmAB_L1eEM26M HLT_g25_medium_tau25_dipion2_tracktwoMVA_50invmAB_L1eEM26M HLT_g25_medium_tau25_dikaonmass_tracktwoMVA_50invmAB_L1eEM26M HLT_g25_medium_tau25_kaonpi1_tracktwoMVA_50invmAB_L1eEM26M HLT_g25_medium_tau25_kaonpi2_tracktwoMVA_50invmAB_L1eEM26M

Trigger Group	HIGP
From 75 bunches	HH 4b jet-seeded 3b1j, Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_3j20c_020jvt_bgn282_pf_ftf_presel2c jet-seeded 2b2j Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn277_pf_ftf_presel2c VBFDelayed stream before Run 474770, Main stream after Run 474770 HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn280_pf_ftf_presel2c muon-seeded 2b2j Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn277_pf_ftf_presel2c VBFDelayed stream before Run 474770, Main stream after Run 474770 HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn280_pf_ftf_presel2c

Trigger Group	Exotics
From 75 bunches	Non-colliding bunches HLT_2g22_tight_L1eEM15_EMPTY HLT_2g22_tight_L1eEM9_UNPAIRED_ISO HLT_2g50_tight_L1eEM15_EMPTY HLT_2g50_tight_L1eEM9_UNPAIRED_ISO HLT_g35_medium_g25_medium_L1eEM15_EMPTY HLT_g35_medium_g25_medium_L1eEM9_UNPAIRED_ISO
From 75 bunches	Dijet + ISR HLT_g85_tight_3j50_L1eEM26M Mono-pole HLT_g0_hiptrt_L1eEM26M LLP, colliding bunches HLT_2mu50_monly_L1MU14FCH HLT_g40_loose_L1eEM24L_mu40_monly_L1MU14FCH HLT_g15_loose_L1eEM10L_2mu10_monly_L12MU8F

Trigger Group	Exotics
	HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_jJ90 HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_jXE80 Non-colliding bunches HLT_g15_loose_L1eEM10L_2mu10_msonly_L1MU3V_EMPTY HLT_g15_loose_L1eEM10L_2mu10_msonly_L1MU3V_UNPAIRED_ISO HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF_EMPTY HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L110DR-MU14FCH-MU5VF_UNPAIRED_ISO HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_EMPTY HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan40_L1MU14FCH_UNPAIRED_ISO

Trigger Group	Supersymmetry
From 75 bunches	B-jet + MET (sbottom) HLT_j75_0eta290_020jvt_bgn260_pf_ftf_xe60_cell_-L12jJ90_jXE80 HLT_j95_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_pfopufit_L1jXE110 HLT_j95_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_tcpufit_L1jXE110 HLT_2j45_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_pfopufit_L12jJ40_jXE110 HLT_2j45_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe85_tcpufit_L12jJ40_jXE110 HLT_3j35_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe70_pfopufit_L13jJ40p0ETA25_jXE80 HLT_3j35_0eta290_020jvt_bgn260_pf_ftf_xe50_-cell_xe70_tcpufit_L13jJ40p0ETA25_jXE80 Photon + multijet (stealth SUSY) HLT_g45_loose_6j45c_L14jJ40p0ETA25

Triggers in 2023

Introduction

Recommended triggers for 2023 physics analyses can be found below. Known issues are detailed in [TriggerIssues2023](#).

See [TriggerNamingRun3](#) for naming conventions.

Trigger periods in intensity ramp-up

Rulebook branches listed are the final tag used for that period, which contain the most stringent rules applied.

Intensity step	Data periods	First run number	Rulebook branch
75 bunches	E1	450360	2023_75b
400 bunches	E2,E3,F3	450894	2023_400b_hybrid
900 bunches	F1,F2,F4	451094	2023_900b_Physics
1200 bunches	F5	451587	RB2023-23.0.26-1200bv2
1800 bunches	F6,F7	451618	RB2023-23.0.26-1800b-CPUv1
2400 bunches	F8,F9,F10	451896	23.0.29

Intensity ramp up after LHC RF issues

Intensity step	Data periods	First run number	Rulebook branch
700 bunches	F11	453319	23.0.29 ↗
1200 bunches	F12	453353	23.0.29 ↗
1800 bunches	F13,F14,F15	453530	23.0.29 ↗
2400 bunches	F16,F17,F18	453733	23.0.29 ↗ (as of June 13)

For runs in each data periods, see COMA [↗](#)

Egamma

Trigger Group	Single electron	Multi electron
75 bunches only	HLT_e26_lhtight_ivarloose_L1EM22VHI HLT_e60_lhmedium_L1EM22VHI HLT_e140_lhloose_L1EM22VHI HLT_e300_etcut_L1EM22VHI HLT_e140_lhloose_noringer_L1EM22VHI HLT_2e24_lhvloose_L12EM20VH Large-radius tracking HLT_e30_lhloose_nopix_lrtmedium_L1EM22VHI	HLT_2e17_lhvloose_L12EM15VHI HLT_2e24_lhvloose_L12EM20VH HLT_e24_lhvloose_2e12_lhvloose_L1EM20VH_3EM1
From 400 bunches	HLT_e26_lhtight_ivarloose_L1eEM26M HLT_e60_lhmedium_L1eEM26M HLT_e140_lhloose_L1eEM26M HLT_e300_etcut_L1eEM26M HLT_e140_lhloose_noringer_L1eEM26M Large-radius tracking HLT_e30_lhloose_nopix_lrtmedium_L1eEM26M	HLT_2e17_lhvloose_L12eEM18M HLT_2e24_lhvloose_L12eEM24L HLT_e24_lhvloose_2e12_lhvloose_L1eEM24L_3eEM1

Trigger Group	Single photon	Multi photon
75 - 900 bunches	HLT_g140_loose_L1EM22VHI HLT_g300_etcut_L1EM22VHI	HLT_2g22_tight_L12EM15VHI HLT_2g50_loose_L12EM20VH HLT_g35_medium_g25_medium_L12EM20VH
From 1200 bunches	HLT_g140_loose_L1eEM26M HLT_g300_etcut_L1eEM26M	HLT_2g20_tight_icaloose_L12eEM18M HLT_2g22_tight_L12eEM18M HLT_2g50_loose_L12eEM24L HLT_g35_medium_g25_medium_L12eEM24L

Trigger Group	Electron + photon
75 - 900 bunches	HLT_e24_lhmedium_g25_medium_02dRAB_L12EM20VH HLT_e24_lhmedium_g12_loose_g12_loose_02dRAB_02dRAC_02dRBC_L1EM20VH_3EM10VH
From 75 bunches	HLT_e24_lhmedium_g25_medium_02dRAB_L12eEM24L HLT_e24_lhmedium_g12_loose_g12_loose_02dRAB_02dRAC_02dRBC_L1eEM24L_3eEM12L

Muon

Trigger Group	Electron + Muon	Photon + Muon
From 400 bunches	N/A	Photon + 2 muon HLT_g35_loose_mu15_mu2noL1_L1eEM28M
From 900 bunches	Electron + muon HLT_e7_lhmedium_L1eEM5_mu24_L1MU14FCH HLT_e17_lhloose_mu14_L1eEM18L_MU8F HLT_e26_lhmedium_mu8noL1_L1eEM26M 2 electron + muon HLT_2e12_lhloose_mu10_L12eEM10L_MU8F Electron + 2muon HLT_e9_lhvloose_mu20_mu8noL1_L1MU14FCH HLT_e12_lhloose_L1eEM10L_2mu10_L12MU8F	Photon + muon HLT_g25_medium_L1eEM18L_mu24_L1MU14FCH HLT_g35_loose_mu18_L1eEM28M HLT_g35_loose_mu15_mu2noL1_L1eEM28M 2 photon + muon HLT_2g10_loose_L1eEM9_mu20_L1MU14FCH

Trigger Group	Single tau	
75 - 1800 bunches	HLT_tau160_mediumRNN_tracktwoMVA_L1TAU100 Long-lived Particle ID	

Trigger Group	Single tau	
	HLT_tau180_mediumRNN_tracktwoLLP_L1TAU100	
From 75 bunches	N/A	HLT_tau35_mediumRNN_tracktwoMVA_tau25_ HLT_tau35_mediumRNN_tracktwoMVA_tau25_ HLT_tau40_mediumRNN_tracktwoMVA_tau35_ HLT_tau80_mediumRNN_tracktwoMVA_tau60_ HLT_tau80_mediumRNN_tracktwoMVA_tau35_ Long-lived Particle ID HLT_tau80_mediumRNN_tracktwoLLP_tau60_m di-tau + MJJ HLT_tau25_mediumRNN_tracktwoMVA_tau20_
From 400 bunches	HLT_tau160_mediumRNN_tracktwoMVA_L1eTAU140	
From 1200 bunches	Long-lived Particle ID HLT_tau180_mediumRNN_tracktwoLLP_L1eTAU140	HLT_tau80_mediumRNN_tracktwoMVA_tau60_ Long-lived Particle ID HLT_tau80_mediumRNN_tracktwoLLP_tau60_m
From 2400 bunches	N/A	Written to VBF Delayed Stream HLT_tau30_mediumRNN_tracktwoMVA_tau20_ HLT_tau30_mediumRNN_tracktwoMVA_tau20_

Electron/Muon + Tau

Trigger Group	Lepton+Tau
75 bunches only	Electron + Tau HLT_e24_lhmedium_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1EM22VHI
From 400 bunches	Electron + Tau HLT_e24_lhmedium_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1eEM26M Muon + Tau HLT_mu20_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1MU14FCH HLT_mu14_ivarloose_tau35_mediumRNN_tracktwoMVA_03dRAB_L1MU8F_TAU20IM HLT_mu14_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1MU8F_TAU12IM_3J12
From 75 bunches	Electron + Tau HLT_e17_lhmedium_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1EM15VHI_2TAU12IM_4

MET (+ X)

Trigger Group	Pure MET	ME

Trigger Group	Pure MET	ME
From 400 bunches	HLT_xe65_cell_xe90_pfopufit_L1XE50 (recommended) HLT_xe80_cell_xe115_tcpufit_L1XE50	HLT_e70_lhloose_xe70_cell_L1eEM26M HLT_g90_loose_xe90_cell_L1eEM26M HLT_mu14_tau25_mediumRNN_tracktwoMVA_xe50_cell_03d HLT_e17_lhmedium_tau25_mediumRNN_tracktwoMVA_xe50 HLT_tau50_mediumRNN_tracktwoMVA_xe80_tcpufit_xe50_ce HLT_tau50_mediumRNN_tracktwoMVA_xe80_pfopufit_xe50_c HLT_tau60_mediumRNN_tracktwoMVA_tau25_mediumRNN_t

Jets

Trigger Group	Single jet	
75 bunches only	HLT_j175f_L1J50p31ETA49	N/A
75, 400 bunches	HLT_j420_L1J100 HLT_j420_ftf_preselj225_L1J100 Large-R HLT_j360_a10t_lcw_jes_L1J100 HLT_j360_a10t_lcw_jes_L1SC111-CJ15 HLT_j420_a10sd_cssk_pf_jes_ftf_preselj225_L1J100	HLT_2j250c_j120c_ftf_presel2j1 HLT_3j200_L1J100 HLT_3j200_ftf_presel3j150_L1J HLT_4j115_ftf_presel4j85_L13J HLT_5j70c_L14J15 HLT_5j70c_ftf_presel5j50_L14J HLT_5j85_L14J15 HLT_5j85_ftf_presel5j50_L14J1 HLT_6j55c_L14J15 HLT_6j55c_ftf_presel6j40_L14J HLT_6j70_L14J15 HLT_6j70_ftf_presel6j40_L14J1 HLT_7j45_L14J15 HLT_7j45_ftf_presel7j30_L14J1 HLT_10j40_L14J15 HLT_10j40_ftf_presel7j30_L14J
From 400 bunches	Small-R HLT_j420_pf_ftf_preselj225_L1J100 From Run 451587: HLT_j400_pf_ftf_preselj225_L1J100 HLT_j220f_L1J75p31ETA49 Large-R HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1J100 HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CJ15 HLT_j460_a10t_lcw_jes_L1J100 HLT_j460_a10t_lcw_jes_L1SC111-CJ15 HLT_j460_a10r_L1J100 HLT_j460_a10r_L1SC111-CJ15 HLT_j460_a10_lcw_subjes_L1J100 HLT_j460_a10_lcw_subjes_L1SC111-CJ15 Large-R with mass cut HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1J100	Small-R HLT_2j250c_j120c_pf_ftf_prese HLT_3j200_pf_ftf_L1J100 HLT_4j115_pf_ftf_presel4j85_L From Run 451587: HLT_4j110_p HLT_4j120_L13J50 HLT_5j70c_pf_ftf_presel5c50_L From Run 451587: HLT_5j65c_p HLT_5j85_pf_ftf_presel5j50_L1 From Run 451587: HLT_5j80_pf From Run 451587: HLT_5j80_pf HLT_6j35c_020jvt_pf_ftf_presel HLT_6j55c_pf_ftf_presel6j40_L HLT_6j70_pf_ftf_presel6j40_L1

Trigger Group	Single jet	
	HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CJ15 HLT_j420_35smcINF_a10t_lcw_jes_L1J100 HLT_j420_35smcINF_a10t_lcw_jes_L1SC111-CJ15 H_T HLT_j0_HT1000_L1HT190-J15s5pETA21 HLT_j0_HT1000_L1J100 HLT_j0_HT1000_pf_ftf_preselcHT450_L1HT190-J15s5pETA21 HLT_j0_HT1000_pf_ftf_preselj180_L1HT190-J15s5pETA21 HLT_j0_HT1000_pf_ftf_preselj180_L1J100	HLT_7j45_pf_ftf_presel7j30_L1 HLT_10j40_pf_ftf_presel7j30_L1 Large-R with mass cut HLT_2j330_35smcINF_a10sd_cssk_pf_jes_ftf_presel2j330_L1 HLT_2j330_35smcINF_a10sd_cssk_pf_jes_ftf_presel2j330_L1 HLT_2j330_35smcINF_a10t_lcw_jes_L1J100 HLT_2j330_35smcINF_a10t_lcw_jes_L1J100 HLT_j360_60smcINF_j360_a10t_lcw_jes_L1J100 HLT_j360_60smcINF_j360_a10t_lcw_jes_L1J100 HLT_j370_35smcINF_j370_a10t_lcw_jes_L1J100 HLT_j370_35smcINF_j370_a10t_lcw_jes_L1J100

+ Delayed stream

B-jets

Trigger Group	Single b-jet
From 75 bunches	GN1 HLT_j225_0eta290_020jvt_bgn170_pf_ftf_preselj180_L1J100 HLT_j300_0eta290_020jvt_bgn177_pf_ftf_preselj225_L1J100 HLT_j360_0eta290_020jvt_bgn185_pf_ftf_preselj225_L1J100
From 1200 bunches	GN1 HLT_j210_0eta290_020jvt_bgn170_pf_ftf_preselj180_L1J100 HLT_j280_0eta290_020jvt_bgn177_pf_ftf_preselj225_L1J100 HLT_j340_0eta290_020jvt_bgn185_pf_ftf_preselj225_L1J100

Trigger Group	Multi b-jet
From 75 bunches	2b HLT_j175_0eta290_020jvt_bgn160_j60_0eta290_020jvt_bgn160_pf_ftf_preselj140b85XXj45b85_L1J100 2b + 1j HLT_j150_2j55_0eta290_020jvt_bgn170_pf_ftf_preselj80XX2j45b90_L1J85_3J30 From Run 451587: HLT_j140_2j50_0eta290_020jvt_bgn170_pf_ftf_preselj80XX2j45b90_L1J85_3J30 3b HLT_3j65_0eta290_020jvt_bgn177_pf_ftf_presel3j45b95_L13J35p0ETA23 From Run 451587: HLT_3j60_0eta290_020jvt_bgn177_pf_ftf_presel3j45b95_L13J35p0ETA23 1b + 3j HLT_j75_0eta290_020jvt_bgn160_3j75_pf_ftf_preselj50b85XX3j50_L14J20 From Run 451587: HLT_j70_0eta290_020jvt_bgn160_3j70_pf_ftf_preselj50b85XX3j50_L14J20 2b + 2j, legacy HH HLT_2j35c_020jvt_bgn160_2j35c_020jvt_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25 2b + 2j

Trigger Group	Multi b-jet
	HLT_2j45_0eta290_020jvt_bgn160_2j45_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25
	2b + HT
	HLT_2j45_0eta290_020jvt_bgn170_j0_HT300_j0_DJMASS700j35_pf_ftf_L1HT150-J20s5pETA31_MJJ-4
	From Run 451587:
	HLT_2j45_0eta290_020jvt_bgn170_j0_HT290_j0_DJMASS700j35_pf_ftf_L1HT150-J20s5pETA31_MJJ-4
	3b + 1j
	HLT_3j35_0eta290_020jvt_bgn170_j35_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25
	4b
	HLT_2j35_0eta290_020jvt_bgn170_2j35_0eta290_020jvt_bgn185_pf_ftf_presel4j25b95_L14J15p0ETA25
	HLT_4j35_0eta290_020jvt_bgn177_pf_ftf_presel4j25b95_L14J15p0ETA25
	2b + 3j
	HLT_2j35_0eta290_020jvt_bgn160_3j35_pf_ftf_presel3j25XX2j25b85_L15J15p0ETA25
	Delayed stream, 1b+5j
	HLT_5j35c_020jvt_j25c_020jvt_SHARED_j25c_020jvt_bgn160_pf_ftf_presel5c25XXc25b85_L14J15

B-physics

Trigger Group	K-star ee Triggers
From 75, 400 bunches	HLT_e5_lhvloose_bBeeM6000_2mu4_l2io_L1BPH-0DR3-EM7J15_2MU3V HLT_e5_lhvloose_bBeeM6000_mu6_l2io_L1BPH-0DR3-EM7J15_MU5VF HLT_e9_lhvloose_e5_lhvloose_bBeeM6000_mu6_l2io_L1BPH-0M9-EM7-EM5_MU5VF HLT_e5_lhvloose_bBeeM6000_L1BKeePrimary HLT_2e5_lhvloose_bBeeM6000_L1BKeePrimary HLT_e5_lhvloose_bBeeM6000_L1BKeePrescaled HLT_2e5_lhvloose_bBeeM6000_L1BKeePrescaled HLT_e9_lhvloose_e5_lhvloose_bBeeM6000_2mu4_l2io_L1BPH-0M9-EM7-EM5_2MU3V

Trigger Group	B-physics muon triggers, no L1Topo
75 bunches only	Dimuon HLT_2mu4_bBmumux_BcmumuD0Xloose_L12MU3V HLT_2mu4_bBmumux_BcmumuDp0loose_L12MU3V HLT_2mu4_bBmumux_BcmumuDsloose_L12MU3V HLT_2mu4_bBmumux_BcmumuDstarloose_L12MU3V HLT_2mu4_bBmumux_BcmumuPi_L12MU3V HLT_2mu4_bBmumux_BdmumuKst_L12MU3V HLT_2mu4_bBmumux_BpmumuKp_L12MU3V HLT_2mu4_bBmumux_BsmumuPhi_L12MU3V HLT_2mu4_bBmumux_LbPqKm_L12MU3V HLT_2mu4_bDimu_L12MU3V HLT_2mu4_bJpsimumu_L12MU3V HLT_2mu4_bUpsimumu_L12MU3V HLT_mu6_mu4_bBmumu_L1MU5VF_2MU3V HLT_mu6_mu4_bBmumu_Lxy0_L1MU5VF_2MU3V HLT_mu6_mu4_bJpsimumu_Lxy0_L1MU5VF_2MU3V HLT_2mu6_bPhi_L12MU5VF

Trigger Group	B-physics muon triggers, no L1Topo
75, 400 bunches	Dimuon
	HLT_2mu4_bBmumux_BcmumuD0Xloose_L12MU3VF
	HLT_2mu4_bBmumux_BcmumuDploose_L12MU3VF
	HLT_2mu4_bBmumux_BcmumuDsloose_L12MU3VF
	HLT_2mu4_bBmumux_BcmumuDstarloose_L12MU3VF
	HLT_2mu4_bBmumux_BcmumuPi_L12MU3VF
	HLT_2mu4_bBmumux_BdmumuKst_L12MU3VF
	HLT_2mu4_bBmumux_BpmumuKp_L12MU3VF
	HLT_2mu4_bBmumux_BsmumuPhi_L12MU3VF
	HLT_2mu4_bBmumux_LbPqKm_L12MU3VF
	HLT_2mu4_bDimu_L12MU3VF
	HLT_2mu4_bJpsimumu_L12MU3VF
	HLT_2mu4_bUpsimumu_L12MU3VF
	HLT_2mu6_bBmumu_L12MU5VF
	HLT_2mu6_bBmumu_Lxy0_L12MU5VF
	HLT_2mu6_bJpsimumu_Lxy0_L12MU5VF
	Bmux
	HLT_mu20_bBmux_BdmuDpX_L1MU14FCH
	HLT_mu20_bBmux_BdmuDstarX_L1MU14FCH
	HLT_mu20_bBmux_BpmuD0X_L1MU14FCH
	HLT_mu20_bBmux_BsmuDsX_L1MU14FCH
	HLT_mu20_bBmux_LbmuLcX_L1MU14FCH
From 75 bunches	Dimuon
	HLT_2mu10_bJpsimumu_L12MU8F
	HLT_2mu10_bUpsimumu_L12MU8F
	HLT_mu11_mu6_bBmumu_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuD0Xloose_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuDploose_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BcmumuPi_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BdmumuKst_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BpmumuKp_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bBmumux_LbPqKm_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bDimu2700_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bDimu_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bDimu_Lxy0_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bJpsimumu_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bJpsimumu_Lxy0_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bPhi_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bTau_L1MU8VF_2MU5VF
	HLT_mu11_mu6_bUpsimumu_L1MU8VF_2MU5VF
	HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF
	HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF
	HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF
	HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF
	HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF
	Multi-muon
	HLT_3mu4_b3mu_noos_L13MU3VF

Trigger Group	B-physics muon triggers, no L1Topo
	HLT_3mu4_bJpsi_L13MU3VF HLT_3mu4_bPhi_L13MU3VF HLT_3mu4_bTau_L13MU3VF HLT_4mu4_bDimu6000_L14MU3V Multiple-candidate L1 flag HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU10BOM HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU12BOM

VBFB

Trigger Group	B-jet + X
From 75 bunches	B-jet + forward jet HLT_j55_0eta290_020jvt_bgn170_2j45f_pf_ftf_preselj45XX2f40_L1J25p0ETA23_2J15p31ETA49 HLT_j80_0eta290_020jvt_bgn170_j60_0eta290_020jvt_bgn185_j45f_pf_ftf_preselj60XXj45XXf40_L1J40p0ETA HLT_j80c_j60_j45f_SHARED_2j45_0eta290_020jvt_bgn160_pf_ftf_preselc60XXj45XXf40_L1J40p0ETA Photon + (b-)jet HLT_g25_medium_4j35a_j0_DJMASS1000j35_L1EM22VHI HLT_g20_tight_j35_0eta290_020jvt_bgn177_3j35a_j0_DJMASS500j35_pf_ftf_L1EM18VHI_MJJ-300 HLT_g25_tight_2j35_0eta290_020jvt_bgn177_-2j35a_pf_ftf_presel2a20b90XX2a20_L1EM22VHI HLT_g25_tight_j35_0eta290_020jvt_bgn177_3j35a_j0_-DJMASS700j35_pf_ftf_-presela20b85XX3a20_-
From 1200 bunches	B-jet + forward jet HLT_j50_0eta290_020jvt_bgn170_2j45f_pf_ftf_preselj45XX2f40_L1J25p0ETA23_2J15p31ETA49 HLT_j80_0eta290_020jvt_bgn170_j60_0eta290_020jvt_bgn185_j45f_pf_ftf_preselj60XXj45XXf40_L1J40p0ETA HLT_j75c_j55_j45f_SHARED_2j45_0eta290_020jvt_bgn160_pf_ftf_preselc60XXj45XXf40_L1J40p0ETA Photon + (b-)jet HLT_g25_medium_4j35a_j0_DJMASS1000j35_L1eEM26M HLT_g20_tight_j35_0eta290_020jvt_bgn177_3j35a_j0_DJMASS500j35_pf_ftf_L1EM18VHI_MJJ-300 HLT_g25_tight_2j35_0eta290_020jvt_bgn177_-2j35a_pf_ftf_presel2a20b90XX2a20_L1eEM26M HLT_g25_tight_j35_0eta290_020jvt_bgn177_3j35a_j0_-DJMASS700j35_pf_ftf_-presela20b85XX3a20_-

Trigger Group	Dijet mass L1
From 75 bunches	Main stream HLT_e10_lhmedium_ivarloose_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF HLT_mu10_ivarmedium_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF HLT_tau25_mediumRNN_tracktwoMVA_tau20_mediumRNN_tracktwoMVA_03dRAB_j70_j50a_j0_DJM Delayed stream HLT_j70_j50a_j0_DJMASS1000j50dphi200x400deta_L1MJJ-500-NFF HLT_2e5_lhmedium_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF HLT_2mu6_2j50a_j0_DJMASS900j50_L1MJJ-500-NFF HLT_j70_j50a_j0_DJMASS1000j50dphi240_xe90_tcpufit_xe50_cell_L1MJJ-500-NFF HLT_j70a_j50a_2j35a_SHARED_2j35_0eta290_020jvt_bgn170_j0_DJMASS1000j50_pf_ftf_presela60XX From Run 451587: HLT_j65a_j45a_2j35a_SHARED_2j35_0eta290_020jvt_bgn170_j0_DJMASS1000j50_pf_ftf_presela60XX HLT_mu4_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1MJJ-500-NFF

Unconventional tracking (LLP)

Trigger Group	Unconventional tracking
75, 400 bunches	Hit-based displaced vertex HLT_j180_hitdvjet200_medium_L1XE50 HLT_j180_hitdvjet260_medium_L1J100 HLT_xe80_tcpufit_hitdvjet200_medium_L1XE50
	High dE/dX HLT_xe80_tcpufit_dedxtrk25_medium_L1XE50
	Disappearing track HLT_distrk20_tight_L1XE50
	Isolated track HLT_xe80_tcpufit_isotrk100_medium_iaggrmedium_L1XE50 HLT_xe80_tcpufit_isotrk120_medium_iaggrloose_L1XE50
From 75 bunches	Hit-based displaced vertex HLT_j180_hitdvjet260_tight_L1J100 HLT_xe80_tcpufit_hitdvjet200_tight_L1XE50
	High dE/dX HLT_xe80_tcpufit_dedxtrk50_medium_L1XE50
	Disappearing track HLT_xe80_tcpufit_distrk20_medium_L1XE50 HLT_xe80_tcpufit_distrk20_tight_L1XE50
	Isolated track HLT_xe80_tcpufit_isotrk120_medium_iaggrmedium_L1XE50 HLT_xe80_tcpufit_isotrk140_medium_iaggrmedium_L1XE50
	Displaced Jet HLT_j180_2dispjet50_3d2p_L1J100 HLT_j180_dispjet50_3d2p_dispjet50_1p_L1J100

Dedicated physics triggers

Trigger Group	Standard Model
75 bunches only	Low-mass Drell-Yan HLT_mu4_ivarloose_mu4_11invAB60_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_7invAB9_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b11invAB60vtx20_L1DY-BOX-2MU3VF HLT_mu4_ivarloose_mu4_b7invAB9vtx20_L1DY-BOX-2MU3VF
75, 400 bunches	HLT_mu6_ivarloose_mu6_11invAB24_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_24invAB60_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b11invAB24vtx20_L1DY-BOX-2MU5VF HLT_mu6_ivarloose_mu6_b24invAB60vtx20_L1DY-BOX-2MU5VF

Trigger Group	Higgs
75 bunches only	Merged electron HLT_e25_mergedtight_g35_medium_90invAB_02dRAB_L12EM20VH

Trigger Group	Higgs
	Photon+meson HLT_g35_medium_tau25_dipion3_tracktwoMVA_L1TAU12_60invAB_L1EM22VHI HLT_g25_medium_tau25_singlepion_tracktwoMVA_50invAB_L1EM22VHI HLT_g25_medium_tau25_dipion1_tracktwoMVA_50invAB_L1EM22VHI HLT_g25_medium_tau25_dipion2_tracktwoMVA_50invAB_L1EM22VHI HLT_g25_medium_tau25_dikaonmass_tracktwoMVA_50invAB_L1EM22VHI HLT_g25_medium_tau25_kaonpi1_tracktwoMVA_50invAB_L1EM22VHI HLT_g25_medium_tau25_kaonpi2_tracktwoMVA_50invAB_L1EM22VHI

From 400 bunches	Merged electron HLT_e25_mergedtight_g35_medium_90invAB_02dRAB_L12eEM24L Photon+meson HLT_g35_medium_tau25_dipion3_tracktwoMVA_L1TAU12_60invAB_L1eEM26M HLT_g25_medium_tau25_singlepion_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dipion1_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dipion2_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_dikaonmass_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_kaonpi1_tracktwoMVA_50invAB_L1eEM26M HLT_g25_medium_tau25_kaonpi2_tracktwoMVA_50invAB_L1eEM26M
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Trigger Group	HDBS
From 75 bunches	HH 4b Main stream HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_3j20c_020jvt_bgn182_pf_ftf_presel2c Delayed stream, jet-seeded HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn177_pf_ftf_presel2c Delayed stream, muon-seeded HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn177_pf_ftf_presel2c
From 1200 bunches	HH 4b Main stream HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_3j20c_020jvt_bgn182_pf_ftf_presel2c Delayed stream, jet-seeded HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn177_pf_ftf_presel2c Delayed stream, muon-seeded HLT_j75c_020jvt_j50c_020jvt_j25c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bgn177_pf_ftf_presel2c

Trigger Group	Exotics
75 bunches only	Non-colliding bunches HLT_2g22_tight_L1EM7_EMPTY HLT_2g22_tight_L1EM7_UNPAIRED_ISO HLT_2g50_tight_L1EM7_EMPTY HLT_2g50_tight_L1EM7_UNPAIRED_ISO HLT_g35_medium_g25_medium_L1EM7_EMPTY HLT_g35_medium_g25_medium_L1EM7_UNPAIRED_ISO
From 400 bunches	Non-colliding bunches HLT_2g22_tight_L1eEM9_EMPTY HLT_2g22_tight_L1eEM9_UNPAIRED_ISO HLT_2g50_tight_L1eEM9_EMPTY HLT_2g50_tight_L1eEM9_UNPAIRED_ISO HLT_g35_medium_g25_medium_L1eEM9_EMPTY

Trigger Group	Exotics
	HLT_g35_medium_g25_medium_L1eEM9_UNPAIRED_ISO
From 75 bunches	Dijet + ISR HLT_g85_tight_3j50_L1eEM26M LLP, colliding bunches HLT_2mu50_msonly_L1MU14FCH HLT_g40_loose_mu40_msonly_L1MU14FCH HLT_g15_loose_2mu10_msonly_L12MU8F Non-colliding bunches HLT_g15_loose_2mu10_msonly_L1MU3V_EMPTY HLT_g15_loose_2mu10_msonly_L1MU3V_UNPAIRED_ISO

Trigger Group	Supersymmetry
From 75 bunches	B-jet + MET (sbottom) HLT_j80_0eta290_020jvt_bgn160_pf_ftf_xe60_cell_L12J50_XE40 + From Run 451587: HLT_j75_0eta290_020jvt_bgn160_pf_ftf_xe60_cell_L12J50_XE40 + HLT_j100_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_pfopufit_L1XE55 From Run 451587: HLT_j95_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_pfopufit_L1XE55 HLT_j100_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_tcpufit_L1XE55 From Run 451587: HLT_j95_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_tcpufit_L1XE55 HLT_2j45_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_pfopufit_L12J15_XE55 HLT_2j45_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe85_tcpufit_L12J15_XE55 HLT_3j35_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe70_pfopufit_L13J15p0ETA25_XE40 HLT_3j35_0eta290_020jvt_bgn160_pf_ftf_xe50_cell_xe70_tcpufit_L13J15p0ETA25_XE40 Photon + multijet (stealth SUSY) HLT_g45_loose_6j45c_L14J15p0ETA25

Triggers in 2022

Introduction

Recommended triggers for 2022 physics analyses can be found below. Known issues are detailed in TriggerIssues2022.

See TriggerNamingRun3 for naming conventions.

Trigger periods in intensity ramp-up

Rulebook branches listed are the final tag used for that period, which contain the most stringent rules applied.

Intensity step	Data periods	First run number	Rulebook branch
75 bunches	Partway through D1	427882	75b_300b_PhysicsP1 ↗
300 bunches	D3,D4	428071	75b_300b_PhysicsP1 ↗
600 bunches	E1,E3	428648	600b-AFP ↗
900 bunches	E4,E5	429018	22.0.77-900b-newStartup ↗

Intensity step	Data periods	First run number	Rulebook branch
1200 bunches	E6	429612	22.0.77-900b-newStartup ↗
1500 bunches	E7	429782	22.0.77-900b-newStartup ↗
1900 bunches	E8	430178	22.0.77-2127b ↗
2100 bunches	--	TBD	22.0.77-2127b ↗

For runs in each data periods, see COMA [↗](#)

Periods D2 and E2 correspond to low-mu operations, so the corresponding run ranges (427929, 428700-428770) use different trigger menus.

Triggers are listed only from the enabling of the HLT at 75 bunches. Prior to that, only L1 streamers were active.

Egamma

Trigger Group	Single electron	Multi electron
75 bunches only	HLT_e17_lhvloose_L1EM15VHI HLT_e20_lhvloose_L1EM15VH HLT_e250_etcut_L1EM22VHI	N/A
From 75 bunches	HLT_e26_lhtight_ivarloose_L1EM22VHI HLT_e60_lhmedium_L1EM22VHI HLT_e140_lhloose_L1EM22VHI HLT_e300_etcut_L1EM22VHI Large-radius tracking HLT_e30_lhloose_nopix_lrtmedium_L1EM22VHI	HLT_2e17_lhvloose_L12EM15VHI HLT_2e24_lhvloose_L12EM20VH HLT_e24_lhvloose_2e12_lhvloose_L1EM20VH_3EM1
From 430536	HLT_e140_lhloose_noringer_L1EM22VHI ★	N/A

★ **Note:** HLT_e140_lhloose_noringer_L1EM22VHI was affected by multiple prescale bugs. This trigger was mostly unprescaled starting from the 75b fills but only reliably unprescaled from run 430896

Trigger Group	Single photon	Multi photon
75 bunches only	HLT_g250_etcut_L1EM22VHI	
From 75 bunches	HLT_g140_loose_L1EM22VHI HLT_g300_etcut_L1EM22VHI	HLT_2g20_tight_icaloose_L12EM15VHI HLT_2g22_tight_L12EM15VHI HLT_2g50_loose_L12EM20VH HLT_g35_medium_g25_medium_L12EM20VH

Trigger Group	Electron + photon
From 75 bunches	HLT_e24_lhmedium_g25_medium_02dRAB_L12EM20VH HLT_e24_lhmedium_g12_loose_g12_loose_02dRAB_02dRAC_02dRBC_L1EM20VH_3EM10VH

Muon

Trigger Group	Single muon	Multi muon
75 bunches only	N/A	Jpsi HLT_mu20_mu2noL1_invmJPsiOS_L1MU14FCH
From 75 bunches	HLT_mu24_ivarmedium_L1MU14FCH HLT_mu50_L1MU14FCH HLT_mu60_0eta105_msonly_L1MU14FCH HLT_mu60_L1MU14FCH HLT_mu80_msonly_3layersEC_L1MU14FCH Large-radius tracking HLT_mu20_LRT_d0loose_L1MU14FCH	HLT_2mu10_l2mt_L1MU10BOM HLT_2mu14_L12MU8F HLT_mu20_ivarmedium_mu8noL1_L1MU14FCH HLT_mu22_mu8noL1_L1MU14FCH HLT_3mu6_L13MU5VF HLT_3mu6_msonly_L13MU5VF HLT_mu20_2mu4noL1_L1MU14FCH HLT_4mu4_L14MU3V Invariant mass cut HLT_mu10_ivarmedium_mu10_10invmAB70_L12MU8F HLT_mu20_ivarmedium_mu4noL1_10invmAB70_L1MU14FCH

Electron/Photon + Muon

Trigger Group	Electron + Muon	Photon + Muon
From 75 bunches	Electron + muon HLT_e7_lhmedium_mu24_L1MU14FCH HLT_e17_lhloose_mu14_L1EM15VH_MU8F HLT_e26_lhmedium_mu8noL1_L1EM22VHI 2 electron + muon HLT_2e12_lhloose_mu10_L12EM8VH_MU8F Electron + 2muon HLT_e9_lhvloose_mu20_mu8noL1_L1MU14FCH HLT_e12_lhloose_2mu10_L12MU8F	Photon + muon HLT_g25_medium_mu24_L1MU14FCH HLT_g35_loose_mu18_L1EM24VHI ★ 2 photon + muon HLT_2g10_loose_mu20_L1MU14FCH Photon + 2 muon HLT_g35_loose_mu15_mu2noL1_L1EM24VHI ★

★ Photon + 1/2 muon triggers seeded by EM24VHI did not collect the full 2022 luminosity due to high load on the LAr ROSES.

Tau

Trigger Group	Single tau	
From 75 bunches	HLT_tau160_mediumRNN_tracktwoMVA_L1TAU100	HLT_tau35_mediumRNN_tracktwoMVA_tau25_n HLT_tau35_mediumRNN_tracktwoMVA_tau25_n

Trigger Group	Single tau	
	Long-lived Particle ID HLT_tau180_mediumRNN_tracktwoLLP_L1TAU100	HLT_tau40_mediumRNN_tracktwoMVA_tau35_n HLT_tau80_mediumRNN_tracktwoMVA_tau60_n HLT_tau80_mediumRNN_tracktwoMVA_tau35_n Long-lived Particle ID HLT_tau80_mediumRNN_tracktwoLLP_tau60_me

Electron/Muon + Tau

Trigger Group	Lepton+Tau
From 75 bunches	Electron + Tau HLT_e24_lhmedium_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1EM22VHI HLT_e17_lhmedium_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1EM15VHI_2TAU12IM_4 Muon + Tau HLT_mu20_ivarloose_tau20_mediumRNN_tracktwoMVA_03dRAB_L1MU14FCH HLT_mu14_ivarloose_tau35_mediumRNN_tracktwoMVA_03dRAB_L1MU8F_TAU20IM HLT_mu14_ivarloose_tau25_mediumRNN_tracktwoMVA_03dRAB_L1MU8F_TAU12IM_3J12

MET (+ X)

Trigger Group	Pure MET	ME
From 75 bunches	HLT_xe65_cell_xe90_pfopufit_L1XE50 (recommended) HLT_xe80_cell_xe115_tcpufit_L1XE50	HLT_e70_lhloose_xe70_cell_L1EM22VHI HLT_g90_loose_xe90_cell_L1EM22VHI HLT_mu14_tau25_mediumRNN_tracktwoMVA_xe50_cell_03d HLT_e17_lhmedium_tau25_mediumRNN_tracktwoMVA_xe50_ HLT_tau60_mediumRNN_tracktwoMVA_tau25_mediumRNN_t

Jets

Trigger Group	Single jet	
75 bunches only	HLT_j175f_L1J50p31ETA49	N/A
75, 300 bunches	HLT_j420_L1J100 HLT_j420_ftf_preselj225_L1J100 Large-R	HLT_2j250c_j120c_ftf_presel2j1 HLT_3j200_L1J100 HLT_3j200_ftf_presel3j150_L1J HLT_4j115_ftf_presel4j85_L13J

Trigger Group	Single jet	
	HLT_j360_a10t_lcw_jes_L1J100 HLT_j360_a10t_lcw_jes_L1SC111-CJ15 HLT_j420_a10sd_cssk_pf_jes_ftf_preselj225_L1J100	HLT_5j70c_L14J15 HLT_5j70c_ftf_presel5j50_L14J15 HLT_5j85_L14J15 HLT_5j85_ftf_presel5j50_L14J15 HLT_6j55c_L14J15 HLT_6j55c_ftf_presel6j40_L14J15 HLT_6j70_L14J15 HLT_6j70_ftf_presel6j40_L14J15 HLT_7j45_L14J15 HLT_7j45_ftf_presel7j30_L14J15 HLT_10j40_L14J15 HLT_10j40_ftf_presel7j30_L14J15
From 75 bunches	Small-R HLT_j420_pf_ftf_preselj225_L1J100 HLT_j220f_L1J75p31ETA49 Large-R HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1J100 HLT_j460_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CJ15 HLT_j460_a10t_lcw_jes_L1J100 HLT_j460_a10t_lcw_jes_L1SC111-CJ15 HLT_j460_a10r_L1J100 HLT_j460_a10r_L1SC111-CJ15 HLT_j460_a10_lcw_subjes_L1J100 HLT_j460_a10_lcw_subjes_L1SC111-CJ15 Large-R with mass cut HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1J100 HLT_j420_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CJ15 HLT_j420_35smcINF_a10t_lcw_jes_L1J100 HLT_j420_35smcINF_a10t_lcw_jes_L1SC111-CJ15 H_T HLT_j0_HT1000_L1HT190-J15s5pETA21 HLT_j0_HT1000_L1J100 HLT_j0_HT1000_pf_ftf_preselcHT450_L1HT190-J15s5pETA21 HLT_j0_HT1000_pf_ftf_preselj180_L1HT190-J15s5pETA21 HLT_j0_HT1000_pf_ftf_preselj180_L1J100	Small-R HLT_2j250c_j120c_pf_ftf_presel5j50_L14J15 HLT_3j200_pf_ftf_L1J100 HLT_4j115_pf_ftf_presel4j85_L14J15 HLT_4j120_L13J50 HLT_5j70c_pf_ftf_presel5c50_L14J15 HLT_5j85_pf_ftf_presel5j50_L14J15 HLT_6j35c_020jvt_pf_ftf_presel6j40_L14J15 HLT_6j55c_pf_ftf_presel6j40_L14J15 HLT_6j70_pf_ftf_presel6j40_L14J15 HLT_7j45_pf_ftf_presel7j30_L14J15 HLT_10j40_pf_ftf_presel7j30_L14J15 Large-R with mass cut HLT_2j330_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1J100 HLT_2j330_35smcINF_a10sd_cssk_pf_jes_ftf_preselj225_L1SC111-CJ15 HLT_2j330_35smcINF_a10t_lcw_jes_L1J100 HLT_2j330_35smcINF_a10t_lcw_jes_L1SC111-CJ15 HLT_j360_60smcINF_j360_a10t_lcw_jes_L1J100 HLT_j360_60smcINF_j360_a10t_lcw_jes_L1SC111-CJ15 HLT_j370_35smcINF_j370_a10t_lcw_jes_L1J100 HLT_j370_35smcINF_j370_a10t_lcw_jes_L1SC111-CJ15

+ Delayed stream

B-jets

Trigger Group	Single b-jet
From 75 bunches	DL1d HLT_j225_0eta290_020jvt_bdl1d70_pf_ftf_preselj180_L1J100 HLT_j300_0eta290_020jvt_bdl1d77_pf_ftf_preselj225_L1J100

Trigger Group	Single b-jet
	HLT_j360_0eta290_020jvt_bdl1d85_pf_ftf_preselj225_L1J100

Trigger Group	Multi b-jet
From 75 bunches	2b HLT_j175_0eta290_020jvt_bdl1d60_j60_0eta290_020jvt_bdl1d60_pf_ftf_preselj140b85XXj45b85_L1J100
	2b + 1j HLT_j150_2j55_0eta290_020jvt_bdl1d70_pf_ftf_preselj80XX2j45b90_L1J85_3J30
	3b HLT_3j65_0eta290_020jvt_bdl1d77_pf_ftf_presel3j45b95_L13J35p0ETA23
	1b + 3j HLT_j75_0eta290_020jvt_bdl1d60_3j75_pf_ftf_preselj50b85XX3j50_L14J20
	2b + 2j, legacy HH HLT_2j35c_020jvt_bdl1d60_2j35c_020jvt_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25
	2b + 2j HLT_2j45_0eta290_020jvt_bdl1d60_2j45_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25
	2b + HT HLT_2j45_0eta290_020jvt_bdl1d70_j0_HT300_j0_DJMASS700j35_pf_ftf_L1HT150-J20s5pETA31_MJJ-
	3b + 1j HLT_3j35_0eta290_020jvt_bdl1d70_j35_pf_ftf_presel2j25XX2j25b85_L14J15p0ETA25
	4b HLT_2j35_0eta290_020jvt_bdl1d70_2j35_0eta290_020jvt_bdl1d85_pf_ftf_presel4j25b95_L14J15p0ETA25 HLT_4j35_0eta290_020jvt_bdl1d77_pf_ftf_presel4j25b95_L14J15p0ETA25
	2b + 3j HLT_2j35_0eta290_020jvt_bdl1d60_3j35_pf_ftf_presel3j25XX2j25b85_L15J15p0ETA25
	Delayed stream, 1b+5j HLT_5j35c_020jvt_j25c_020jvt_SHARED_j25c_020jvt_bdl1d60_pf_ftf_presel5c25XXc25b85_L14J15

B-physics

Trigger Group	K-star ee Triggers
From 75, 300 bunches	HLT_e5_lhvloose_bBeeM6000_2mu4_l2io_L1BPH-0DR3-EM7J15_2MU3V
	HLT_e5_lhvloose_bBeeM6000_mu6_l2io_L1BPH-0DR3-EM7J15_MU5VF
	HLT_e9_lhvloose_e5_lhvloose_bBeeM6000_mu6_l2io_L1BPH-0M9-EM7-EM5_MU5VF
	HLT_e5_lhvloose_bBeeM6000_L1BKeePrimary 
	HLT_2e5_lhvloose_bBeeM6000_L1BKeePrimary 
	HLT_e5_lhvloose_bBeeM6000_L1BKeePrescaled 
	HLT_2e5_lhvloose_bBeeM6000_L1BKeePrescaled 
	HLT_e9_lhvloose_e5_lhvloose_bBeeM6000_2mu4_l2io_L1BPH-0M9-EM7-EM5_2MU3V 

★ **Note:** These BKEE triggers were prescaled at various points due to high LAr ROS access rates limiting the HLT operation. From Run 431241, they were furthermore run only as End-Of-Fill triggers, after mu had declined during the run.

★ **Note:** The HLT item is unprescaled, but it is fed by L1 items that do have prescales applied

Trigger Group	B-physics muon triggers, no L1Topo
75 bunches only	Dimuon HLT_2mu4_bBmumux_BcmumuD0Xloose_L12MU3V HLT_2mu4_bBmumux_BcmumuDploose_L12MU3V HLT_2mu4_bBmumux_BcmumuDsloose_L12MU3V HLT_2mu4_bBmumux_BcmumuDstarloose_L12MU3V HLT_2mu4_bBmumux_BcmumuPi_L12MU3V HLT_2mu4_bBmumux_BdmumuKst_L12MU3V HLT_2mu4_bBmumux_BpmumuKp_L12MU3V HLT_2mu4_bBmumux_BsmumuPhi_L12MU3V HLT_2mu4_bBmumux_LbPqKm_L12MU3V HLT_2mu4_bDimu_L12MU3V HLT_2mu4_bJpsimumu_L12MU3V HLT_2mu4_bUpsimumu_L12MU3V HLT_mu6_mu4_bBmumu_L1MU5VF_2MU3V HLT_mu6_mu4_bBmumu_Lxy0_L1MU5VF_2MU3V HLT_mu6_mu4_bJpsimumu_Lxy0_L1MU5VF_2MU3V HLT_2mu6_bPhi_L12MU5VF
75, 300 bunches	Dimuon HLT_2mu4_bBmumux_BcmumuD0Xloose_L12MU3VF HLT_2mu4_bBmumux_BcmumuDploose_L12MU3VF HLT_2mu4_bBmumux_BcmumuDsloose_L12MU3VF HLT_2mu4_bBmumux_BcmumuDstarloose_L12MU3VF HLT_2mu4_bBmumux_BcmumuPi_L12MU3VF HLT_2mu4_bBmumux_BdmumuKst_L12MU3VF HLT_2mu4_bBmumux_BpmumuKp_L12MU3VF HLT_2mu4_bBmumux_BsmumuPhi_L12MU3VF HLT_2mu4_bBmumux_LbPqKm_L12MU3VF HLT_2mu4_bDimu_L12MU3VF HLT_2mu4_bJpsimumu_L12MU3VF HLT_2mu4_bUpsimumu_L12MU3VF HLT_2mu6_bBmumu_L12MU5VF HLT_2mu6_bBmumu_Lxy0_L12MU5VF HLT_2mu6_bJpsimumu_Lxy0_L12MU5VF Bmux HLT_mu20_bBmux_BdmuDpX_L1MU14FCH HLT_mu20_bBmux_BdmuDstarX_L1MU14FCH HLT_mu20_bBmux_BpmuD0X_L1MU14FCH HLT_mu20_bBmux_BsmuDsX_L1MU14FCH HLT_mu20_bBmux_LbmuLcX_L1MU14FCH
From 75 bunches	Dimuon HLT_2mu10_bJpsimumu_L12MU8F HLT_2mu10_bUpsimumu_L12MU8F HLT_mu11_mu6_bBmumu_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuD0Xloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDploose_L1MU8VF_2MU5VF

Trigger Group	B-physics muon triggers, no L1Topo
	HLT_mu11_mu6_bBmumux_BcmumuDsloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuDstarloose_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BcmumuPi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BdmumuKst_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BpmumuKp_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_BsmumuPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bBmumux_LbPqKm_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu2700_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_L1MU8VF_2MU5VF HLT_mu11_mu6_bDimu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_L1MU8VF_2MU5VF HLT_mu11_mu6_bJpsimumu_Lxy0_L1MU8VF_2MU5VF HLT_mu11_mu6_bPhi_L1MU8VF_2MU5VF HLT_mu11_mu6_bTau_L1MU8VF_2MU5VF HLT_mu11_mu6_bUpsimumu_L1MU8VF_2MU5VF HLT_mu6_2mu4_bDimu2700_L1MU5VF_3MU3VF HLT_mu6_2mu4_bDimu6000_L1MU5VF_3MU3VF HLT_mu6_2mu4_bJpsi_L1MU5VF_3MU3VF HLT_mu6_2mu4_bTau_L1MU5VF_3MU3VF HLT_mu6_2mu4_bUpsi_L1MU5VF_3MU3VF Multi-muon HLT_3mu4_b3mu_noos_L13MU3VF HLT_3mu4_bJpsi_L13MU3VF HLT_3mu4_bPhi_L13MU3VF HLT_3mu4_bTau_L13MU3VF HLT_4mu4_bDimu6000_L14MU3V Multiple-candidate L1 flag HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU10BOM HLT_mu10_l2mt_mu4_l2mt_bJpsimumu_L1MU12BOM

VBF

Trigger Group	B-jet + X
From 75 bunches	B-jet + forward jet HLT_j55_0eta290_020jvt_bdl1d70_2j45f_pf_ftf_preselj45XX2f40_L1J25p0ETA23_2J15p31ETA49 HLT_j80_0eta290_020jvt_bdl1d70_j60_0eta290_020jvt_bdl1d85_j45f_pf_ftf_preselj60XXj45XXf40_L1J4 HLT_j80c_j60_j45f_SHARED_2j45_0eta290_020jvt_bdl1d60_pf_ftf_preselc60XXj45XXf40_L1J40p0ETA Photon + (b-)jet HLT_g25_medium_4j35a_j0_DJMASS1000j35_L1EM22VHI ★ HLT_g20_tight_icaloose_j35_0eta290_020jvt_bdl1d77_3j35a_j0_DJMASS500j35_pf_ftf_L1EM18VHI HLT_g25_tight_icaloose_2j35_0eta290_020jvt_bdl1d77_2j35a_pf_ftf_L1EM22VHI ★ HLT_g25_tight_icaloose_j35_0eta290_020jvt_bdl1d77_3j35a_j0_DJMASS700j35_pf_ftf_L1EM22VHI ★ HLT_g25_tight_icaloose_2j35_0eta290_020jvt_bdl1d77_2j35a_pf_ftf_presel2a20b90XX2a20_L1EM22V HLT_g25_tight_icaloose_j35_0eta290_020jvt_bdl1d77_3j35a_j0_DJMASS700j35_pf_ftf_presela20b85X

★ Due to limitations from a high ReadOut Server (ROS) rate, the g25 triggers seeded from L1_EM22VHI did not collect the full 2022 luminosity

Trigger Group	Dijet mass L1
From 75 bunches	Main stream
	HLT_e10_lhmedium_ivarloose_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF
	HLT_mu10_ivarmedium_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF
	HLT_tau25_mediumRNN_tracktwoMVA_tau20_mediumRNN_tracktwoMVA_03dRAB_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF
	Delayed stream
	HLT_j70_j50a_j0_DJMASS1000j50dphi200x400deta_L1MJJ-500-NFF
	HLT_2e5_lhmedium_j70_j50a_j0_DJMASS900j50_L1MJJ-500-NFF
	HLT_2mu6_2j50a_j0_DJMASS900j50_L1MJJ-500-NFF
	HLT_j70_j50a_j0_DJMASS1000j50dphi240_xe90_tcpufit_xe50_cell_L1MJJ-500-NFF
	HLT_j70a_j50a_2j35a_SHARED_2j35_0eta290_020jvt_bd11d70_j0_DJMASS1000j50_pf_ftf_presela60XX
	HLT_mu4_j70_j50a_j0_DJMASS1000j50_xe50_tcpufit_L1MJJ-500-NFF

Unconventional tracking (LLP)

Trigger Group	Unconventional tracking
75, 300 bunches	Hit-based displaced vertex
	HLT_hitdvjet200_medium_L1XE50
	HLT_hitdvjet260_medium_L1J100
	HLT_xe80_tcpufit_hitdvjet200_medium_L1XE50
	High dE/dX
	HLT_xe80_tcpufit_dedxtrk25_medium_L1XE50
	Disappearing track
	HLT_distrk20_tight_L1XE50
	Isolated track
	HLT_xe80_tcpufit_isotrk100_medium_iaggrmedium_L1XE50 HLT_xe80_tcpufit_isotrk120_medium_iaggrloose_L1XE50
From 75 bunches	Hit-based displaced vertex
	HLT_hitdvjet260_tight_L1J100 ★
	HLT_xe80_tcpufit_hitdvjet200_tight_L1XE50
	High dE/dX
	HLT_xe80_tcpufit_dedxtrk50_medium_L1XE50
	Disappearing track
	HLT_xe80_tcpufit_distrk20_medium_L1XE50 HLT_xe80_tcpufit_distrk20_tight_L1XE50
	Isolated track
	HLT_xe80_tcpufit_isotrk120_medium_iaggrmedium_L1XE50 HLT_xe80_tcpufit_isotrk140_medium_iaggrmedium_L1XE50

★ Note: due to high CPU cost HLT_hitdvjet260_tight_L1J100 did not collect the full 2022 luminosity

Dedicated physics triggers

Trigger Group	Standard Model
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Trigger Group	Standard Model
75 bunches only	Bell Inequality
	HLT_2mu3_L12MU3V
	Low-mass Drell-Yan
	HLT_mu4_ivarloose_mu4_11invAB60_L12MU3V
	HLT_mu4_ivarloose_mu4_7invAB9_L12MU3V
75, 300 bunches	HLT_mu4_ivarloose_mu4_b11invAB60vtx20_L12MU3V
	HLT_mu4_ivarloose_mu4_b7invAB9vtx20_L12MU3V
	HLT_mu6_ivarloose_mu6_11invAB24_L12MU5VF
	HLT_mu6_ivarloose_mu6_24invAB60_L12MU5VF
	HLT_mu6_ivarloose_mu6_b11invAB24vtx20_L12MU5VF
	HLT_mu6_ivarloose_mu6_b24invAB60vtx20_L12MU5VF

Trigger Group	Higgs
From 75 bunches	Merged electron
	HLT_e25_mergedtight_g35_medium_90invAB_02dRAB_L12EM20VH
	Photon+meson
	HLT_g35_medium_tau25_dipion3_tracktwoMVA_60invAB_L1EM22VHI
	HLT_g25_medium_tau25_singlepion_tracktwoMVA_50invAB_L1EM22VHI ★
	HLT_g25_medium_tau25_dipion1_tracktwoMVA_50invAB_L1EM22VHI ★
	HLT_g25_medium_tau25_dipion2_tracktwoMVA_50invAB_L1EM22VHI ★
	HLT_g25_medium_tau25_dikaonmass_tracktwoMVA_50invAB_L1EM22VHI ★
	HLT_g25_medium_tau25_kaonpi1_tracktwoMVA_50invAB_L1EM22VHI ★
	HLT_g25_medium_tau25_kaonpi2_tracktwoMVA_50invAB_L1EM22VHI ★

★ Due to limitations from a high ReadOut Server (ROS) rate, the g25 triggers seeded from L1_EM22VHI did not collect the full 2022 luminosity

Trigger Group	HDBS
From 75 bunches	HH 4b
	Main stream
	HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_3j20c_020jvt_bdl1d82_pf_ftf_presel2c
	Delayed stream, jet-seeded
	HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bdl1d77_pf_ftf_presel2c
	Delayed stream, muon-seeded
	HLT_j80c_020jvt_j55c_020jvt_j28c_020jvt_j20c_020jvt_SHARED_2j20c_020jvt_bdl1d77_pf_ftf_presel2c

Trigger Group	Exotics
From 75 bunches	Dijet + ISR
	HLT_g85_tight_3j50_L1EM22VHI
	LLP, colliding bunches
	HLT_2mu50_msonly_L1MU14FCH
	HLT_g40_loose_mu40_msonly_L1MU14FCH
	HLT_g15_loose_2mu10_msonly_L12MU8F
	Non-colliding bunches
	HLT_g15_loose_2mu10_msonly_L1MU3V_EMPTY
	HLT_g15_loose_2mu10_msonly_L1MU3V_UNPAIRED_ISO
	HLT_2g22_tight_L1EM7_EMPTY
	HLT_2g22_tight_L1EM7_UNPAIRED_ISO
	HLT_2g50_tight_L1EM7_EMPTY
	HLT_2g50_tight_L1EM7_UNPAIRED_ISO

Trigger Group	Exotics
	HLT_g35_medium_g25_medium_L1EM7_EMPTY HLT_g35_medium_g25_medium_L1EM7_UNPAIRED_ISO

Trigger Group	Supersymmetry
From 75 bunches	B-jet + MET (sbottom) HLT_j80_0eta290_020jvt_bdl1d60_pf_ftf_xe60_cell_L12J50_XE40 + HLT_j100_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe85_pfopufit_L1XE55 HLT_j100_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe85_tcpufit_L1XE55 HLT_2j45_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe85_pfopufit_L12J15_XE55 HLT_2j45_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe85_tcpufit_L12J15_XE55 HLT_3j35_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe70_pfopufit_L13J15p0ETA25_XE40 HLT_3j35_0eta290_020jvt_bdl1d60_pf_ftf_xe50_cell_xe70_tcpufit_L13J15p0ETA25_XE40 Photon + multijet (stealth SUSY) HLT_g45_loose_6j45c_L14J15p0ETA25

★ XE50 cut applied during ramp-up period

Triggers in 2018

Introduction

Recommended triggers for 2018 physics analyses can be found below. Known issues are detailed in TriggerIssues2018.

Data Periods

Some useful COMA ConditionsMetadata links for data18_13TeV are:

- COMA Periods of data18_13TeV [↗](#)
- COMA Period Runs in data18_13TeV [↗](#)
- COMA Period Triggers in data18_13TeV [↗](#)
- COMA Period Streams in data18_13TeV [↗](#)

Rulebook tags

- Tag for AthenaP1,21.1.26 (ATR-17908 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.27 (ATR-18026 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.29 (ATR-18186 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.30 (ATR-18233 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.31 (ATR-18334 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.33 (ATR-18404 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.36 (ATR-18487 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.37 (ATR-18515 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.39 (ATR-18570 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.40 (ATR-18635 [↗](#)), Rulebook [↗](#)
- Tag for AthenaP1,21.1.41 (ATR-18679 [↗](#)), Rulebook [↗](#)

- Tag for AthenaP1,21.1.43 (ATR-18749[↗](#)), Rulebook[↗](#)
- Tag for AthenaP1,21.1.45 (ATR-18840[↗](#)), Rulebook[↗](#)
- Tag for AthenaP1,21.1.48 (ATR-18988[↗](#)), Rulebook[↗](#)
- Tag for AthenaP1,21.1.50 (ATR-19026[↗](#)), Rulebook[↗](#)

Egamma

Trigger Group	Single electron	Multi electron
period B-	HLT_e26_lhtight_nod0_ivarloose HLT_e26_lhtight_nod0* HLT_e60_lhmedium_nod0 HLT_e140_lhloose_nod0 HLT_e300_etcut	HLT_2e17_lhvloose_nod0_L12EM15VHI HLT_2e24_lhvloose_nod0 (L1_2EM20VH) HLT_e24_lhvloose_nod0_2e12_lhvloose_nod0_L1EM20VH_3EM10VH

* Unprescaled since run 349169

Trigger Group	Single photon	Multi photon
period B-	HLT_g140_loose HLT_g300_etcut	HLT_2g20_tight_icalovloose_L12EM15VHI HLT_2g22_tight_L12EM15VHI HLT_2g25_tight_L12EM20VH HLT_g35_medium_g25_medium_L12EM20VH HLT_2g50_loose_L12EM20VH HLT_2g25_loose_g15_loose

Egamma + MET

Trigger Group	electron+MET	photon+MET
period B-	HLT_e70_lhloose_nod0_xe70noL1	HLT_g80_loose_xe80noL1

Muon

Trigger Group	Single muon	Multi muon
period B-	HLT_mu26_ivarmedium HLT_mu50 HLT_mu60_0eta105_msonly	HLT_2mu14 (L1_2MU10) HLT_mu22_mu8noL1 (L1_MU20) HLT_mu20_2mu4noL1 (L1_MU20) HLT_3mu6 (L1_3MU6)

Electron + Muon

	e+mu	e+2mu	2e+mu
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Trigger Group			
period B-	HLT_e17_lhloose_nod0_mu14 HLT_e26_lhmedium_nod0_mu8noL1 HLT_e7_lhmedium_nod0_mu24	HLT_e12_lhloose_nod0_2mu10	HLT_2e12_lhloose_nod0_mu10

Electron/Muon + Photon

Trigger Group	electron+photon	muon+photon	multi-muon
period B-	HLT_e24_lhmedium_nod0_L1EM20VH_g25_medium	HLT_g25_medium_mu24	HLT_g35_loose_L1EM24 HLT_g35_loose_L1EM24 HLT_g35_tight_icalotight HLT_g35_tight_icalotight

Bphysics

Trigger Group	Bphysics
period B-	HLT_2mu6_bBmumu_Lxy0_L1BPH-2M9-2MU6_BPH-2DR15-2MU6 HLT_2mu6_bBmumux_BsmumuPhi_L1BPH-2M9-2MU6_BPH-2DR15-2MU6 Generic HLT_mu11_mu6_bDimu_L1LFV-MU11 HLT_mu11_mu6_bDimu (L1_MU11_2MU6) HLT_mu11_mu6_bDimu_Lxy0_L1LFV-MU11 HLT_mu11_mu6_bDimu_Lxy0 (L1_MU11_2MU6) HLT_mu11_mu6_bDimu2700_L1LFV-MU11 HLT_mu11_mu6_bDimu2700 (L1_MU11_2MU6) HLT_mu11_mu6_bDimu2700_Lxy0_L1LFV-MU11 HLT_mu11_mu6_bDimu2700_Lxy0 (L1_MU11_2MU6) B->mumu HLT_mu11_mu6_bBmumu_L1LFV-MU11 HLT_mu11_mu6_bBmumu (L1_MU11_2MU6) HLT_mu11_mu6_bBmumuxv2_L1LFV-MU11 HLT_mu11_mu6_bBmumuxv2 (L1_MU11_2MU6) HLT_2mu6_bBmumux_BpmumuKp_L1BPH-2M9-2MU6_BPH-2DR15-2MU6 HLT_mu11_mu6_bBmumux_BpmumuKp_L1LFV-MU11 HLT_mu11_mu6_bBmumux_BpmumuKp (L1_MU11_2MU6) Jpsi->mumu HLT_2mu6_bJpsimumu_L1BPH-2M9-2MU6_BPH-2DR15-2MU6 HLT_2mu6_bJpsimumu_Lxy0_L1BPH-2M9-2MU6_BPH-2DR15-2MU6 HLT_mu11_mu6_bJpsimumu_L1LFV-MU11 HLT_mu11_mu6_bJpsimumu (L1_MU11_2MU6) HLT_mu11_mu6_bJpsimumu_Lxy0_L1LFV-MU11 HLT_mu11_mu6_bJpsimumu_Lxy0 (L1_MU11_2MU6)

Trigger Group	Bphysics
	Upsi->mumu HLT_2mu6_bUpsimumu_L1BPH-8M15-2MU6_BPH-0DR22-2MU6* HLT_mu11_mu6_bUpsimumu_L1LFV-MU11 HLT_mu11_mu6_bUpsimumu (L1_MU11_2MU6) B->mumuX HLT_mu11_mu6_bBmumuxv2_L1LFV-MU11 HLT_mu11_mu6_bBmumuxv2 (L1_MU11_2MU6) Tau->3mu HLT_mu11_mu6_bTau_L1LFV-MU11 HLT_mu11_mu6_bTau (L1_MU11_2MU6) HLT_mu11_2mu4noL1_bTau_L1MU11_2MU6 HLT_mu11_2mu2btrk_bTauTight_L1MU11_2MU6** HLT_mu11_2mu4noL1_bNocut_L1MU11_2MU6 HLT_mu20_mu6noL1_bTau (L1_MU20) HLT_mu20_mu6btrk_bTauTight** (L1_MU20) HLT_mu20_2mu2btrk_bTauTight** (L1_MU20) phi->mumu HLT_mu11_mu6_bPhi_L1LFV-MU11 HLT_mu11_mu6_bPhi (L1_MU11_2MU6) HLT_mu11_mu6noL1_bPhi_L1MU11_2MU6 L1_3MU4 HLT_3mu4_bJpsi HLT_3mu4_bUpsi HLT_3mu4_bTau HLT_3mu4_bPhi L1_MU6_3MU4 HLT_mu6_2mu4_bDimu2700 L1_3MU6 HLT_3mu6_bDimu L1_4MU4 HLT_4mu4

* Included in Main and not in BphysLS stream ** Unprescaled since run 350067

Trigger Group	B->K*ee
period F	HLT_e9_lhloose_e5_lhloose_bBeexM6000_2mu4_nomucomb_L1BPH-0M9-EM7-EM5_2MU4
period K-	HLT_e9_lhvloose_nod0_e5_lhvloose_nod0_bBeexM6000t_2mu4_nomucomb_L1BPH-0M9-EM7-EM5_2MU4

Taus

Trigger Group	taus
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Trigger Group	taus
period	single tau
B-	HLT_tau160_medium1_tracktwoEF_L1TAU100
	single tau (backup)
	HLT_tau200_medium1_tracktwoEF_L1TAU100
	di-tau
	HLT_tau80_medium1_tracktwoEF_L1TAU60_tau60_medium1_tracktwoEF_L1TAU40 (L1_TAU60_2TAU40)
	di-tau (L1Topo)
	HLT_tau80_medium1_tracktwoEF_L1TAU60_tau35_medium1_tracktwoEF_L1TAU12IM_L1TAU60_DR-TAU12I-J25
	di-tau+1J
	HLT_tau40_medium1_tracktwoEF_tau35_medium1_tracktwoEF (L1_TAU25IM_2TAU20IM_2J25_3J20)
	di-tau+1J (L1Topo)
	HLT_tau35_medium1_tracktwoEF_tau25_medium1_tracktwoEF_L1DR-TAU20ITAU12I-J25
	di-tau+1J (L1Topo) (backup)
	HLT_tau35_medium1_tracktwoEF_tau25_medium1_tracktwoEF_03dR30_L1DR-TAU20ITAU12I-J25
period	single tau
K-	HLT_tau160_mediumRNN_tracktwoMVA_L1TAU100
	single tau (backup)
	HLT_tau200_mediumRNN_tracktwoMVA_L1TAU100
	di-tau
	HLT_tau80_mediumRNN_tracktwoMVA_L1TAU60_tau60_mediumRNN_tracktwoMVA_L1TAU40 (L1_TAU60_2TAU40)
	di-tau (L1Topo)
	HLT_tau80_mediumRNN_tracktwoMVA_L1TAU60_tau35_mediumRNN_tracktwoMVA_L1TAU12IM_L1TAU60_DR-TAU12I-J25
	di-tau+1J
	HLT_tau40_mediumRNN_tracktwoMVA_tau35_mediumRNN_tracktwoMVA (L1_TAU25IM_2TAU20IM_2J25_3J20)
	di-tau+1J (L1Topo)
	HLT_tau35_mediumRNN_tracktwoMVA_tau25_mediumRNN_tracktwoMVA_L1DR-TAU20ITAU12I-J25
	di-tau+1J (L1Topo) (backup)
	HLT_tau35_mediumRNN_tracktwoMVA_tau25_mediumRNN_tracktwoMVA_03dR30_L1DR-TAU20ITAU12I-J25

Taus + electron/muon

Trigger Group	tau+electron
period	tau+e+1J
B-	HLT_e17_lhmedium_nod0_ivarloose_tau25_medium1_tracktwoEF (L1_EM15VHI_2TAU12IM_J25_3J12)
	tau+e+1J (backup)
	HLT_e24_lhmedium_nod0_ivarloose_tau35_medium1_tracktwoEF (L1_EM20VHI_TAU20IM_2TAU20_J25_3J20)

Trigger Group	tau+electron
	tau+e+1J (L1Topo) HLT_e17_lhmedium_nod0_ivarloose_tau25_medium1_tracktwoEF_L1DR-EM15TAU12I-J25 tau+e+2J HLT_e17_lhmedium_nod0_ivarloose_tau25_medium1_tracktwoEF_L1EM15VHI_2TAU12IM_4J12
period K-	tau+e+1J HLT_e17_lhmedium_nod0_ivarloose_tau25_mediumRNN_tracktwoMVA (L1_EM15VHI_2TAU12IM_J25_3J12) tau+e+1J (backup) HLT_e24_lhmedium_nod0_ivarloose_tau35_mediumRNN_tracktwoMVA (L1_EM20VHI_TAU20IM_2TAU20_J25_3J20) tau+e+1J (L1Topo) HLT_e17_lhmedium_nod0_ivarloose_tau25_mediumRNN_tracktwoMVA_L1DR-EM15TAU12I-J25 tau+e+2J HLT_e17_lhmedium_nod0_ivarloose_tau25_mediumRNN_tracktwoMVA_L1EM15VHI_2TAU12IM_4J12

Taus + MET + electron/muon

Trigger Group	tau+MET+electron/muon
period B-	tau+xe+1J HLT_tau35_medium1_tracktwoEF_xe70_L1XE45 (L1_TAU20IM_2J20_XE45) di-tau+xe HLT_tau60_medium1_tracktwoEF_tau25_medium1_tracktwoEF_xe50 (L1_TAU40_2TAU12IM_XE40) tau+electron+xe HLT_e17_lhmedium_nod0_tau25_medium1_tracktwoEF_xe50 (L1_EM15VHI_2TAU12IM_XE35) tau+electron+xe (backup) HLT_e17_lhmedium_nod0_ivarloose_tau25_medium1_tracktwoEF_xe50 (L1_EM15VHI_2TAU12IM_XE35) tau+muon+xe HLT_mu14_tau25_medium1_tracktwoEF_xe50 (L1_MU10_TAU12IM_XE35) tau+muon+xe (backup) HLT_mu14_ivarloose_tau25_medium1_tracktwoEF_xe50 (L1_MU10_TAU12IM_XE35)
period K-	tau+xe+1J HLT_tau35_mediumRNN_tracktwoMVA_xe70_L1XE45 (L1_TAU20IM_2J20_XE45) di-tau+xe

Trigger Group	tau+MET+electron/muon
	HLT_tau60_mediumRNN_tracktwoMVA_tau25_mediumRNN_tracktwoMVA_xe50 (L1_TAU40_2TAU12IM_XE40) tau+electron+xe HLT_e17_lhmedium_nod0_tau25_mediumRNN_tracktwoMVA_xe50 (L1_EM15VHI_2TAU12IM_XE35) tau+electron+xe (backup) HLT_e17_lhmedium_nod0_ivarloose_tau25_mediumRNN_tracktwoMVA_xe50 (L1_EM15VHI_2TAU12IM_XE35) tau+muon+xe HLT_mu14_tau25_mediumRNN_tracktwoMVA_xe50 (L1_MU10_TAU12IM_XE35) tau+muon+xe (backup) HLT_mu14_ivarloose_tau25_mediumRNN_tracktwoMVA_xe50 (L1_MU10_TAU12IM_XE35)

Jets

Trigger Group	Single jet	Multi jets
period	small-R	small-R
B-	HLT_j420 HLT_j225_gsc420_boffperf_split HLT_j260_320eta490 large-R HLT_j460_a10r_L1SC111 HLT_j460_a10r_L1J100 HLT_j460_a10_lcw_subjes_L1SC111 HLT_j460_a10_lcw_subjes_L1J100 HLT_j460_a10t_lcw_jes_L1SC111 HLT_j460_a10t_lcw_jes_L1J100 large-R with mass cut HLT_j420_a10t_lcw_jes_35smcINF_L1SC111* HLT_j420_a10t_lcw_jes_35smcINF_L1J100	HLT_3j200 HLT_4j120 HLT_4j85_gsc115_boffperf_split HLT_5j50_gsc70_boffperf_split_0eta240_L14J15 HLT_5j60_gsc85_boffperf_split_L14J15 HLT_5j70_0eta240_L14J15 HLT_5j85_L14J15 HLT_6j45_gsc55_boffperf_split_0eta240_L14J15 HLT_6j50_gsc70_boffperf_split_L14J15 HLT_6j55_0eta240_L14J15 HLT_6j70_L14J15 HLT_7j35_gsc45_boffperf_split_L14J15 HLT_7j45_L14J15 HLT_10j40_L14J15 large-R with mass cut HLT_2j330_a10t_lcw_jes_35smcINF_L1SC111 HLT_2j330_a10t_lcw_jes_35smcINF_L1J100 HLT_j360_a10t_lcw_jes_60smcINF_j360_a10t_lcw_jes_L14J15 HLT_j370_a10t_lcw_jes_35smcINF_j370_a10t_lcw_jes_L14J15

* Included since run 349498

MET

Trigger Group	MET
period B	HLT_xe110_pufit_xe70_L1XE50 HLT_xe120_pufit_L1XE50
period C-J	HLT_xe110_pufit_xe65_L1XE50* HLT_xe110_pufit_xe70_L1XE50 HLT_xe120_pufit_L1XE50
period K-	HLT_xe110_pufit_xe65_L1XE50 HLT_xe110_pufit_xe70_L1XE50

* Unprescaled since run 350067

HT

Trigger Group	HT
period B-	HLT_ht1000_L1J100

Bjet

Trigger Group	Single bjet	Multi bjet
period B-	1b HLT_j175_gsc225_bmv2c1040_split HLT_j225_gsc275_bmv2c1060_split HLT_j225_gsc300_bmv2c1070_split HLT_j225_gsc360_bmv2c1077_split 1b+3j HLT_j55_gsc75_bmv2c1040_split_3j55_gsc75_boffperf_split HLT_j60_gsc85_bmv2c1050_split_3j60_gsc85_boffperf_split	2b HLT_j150_gsc175_bmv2c1060_split_j45_gsc 2b+1j HLT_j110_gsc150_boffperf_split_2j45_gsc55 2b+2j HLT_2j35_bmv2c1060_split_2j35_L14J15.0E HLT_2j35_gsc45_bmv2c1050_split_2j35_gsc HLT_2j45_gsc55_bmv2c1060_split_2j45_gsc 2b+3j HLT_2j35_bmv2c1060_split_3j35 HLT_2j35_gsc45_bmv2c1060_split_3j35_gsc 3b HLT_3j50_gsc65_bmv2c1077_split_L13J35.0 3b+1j HLT_3j35_bmv2c1070_split_j35_L14J15.0E 4b HLT_4j35_bmv2c1077_split_L14J15.0ETA2 HLT_2j35_bmv2c1070_split_2j35_bmv2c108
period L-	1b* HLT_j225_gsc275_bhmv2c1060_split*	

Trigger Group	Single bjet	Multi bjet
	HLT_j225_gsc300_bhm2c1070_split* HLT_j225_gsc360_bhm2c1077_split	

* Unprescaled since run 358541

Bjet + HT

Trigger Group	Bjet+HT
period B-	1b+HT HLT_j45_gsc55_bmv2c1050_split_ht700_L1HT190-J15s5.ETA21 2b+HT HLT_2j45_gsc55_bmv2c1050_split_ht300_L1HT190-J15s5.ETA21

VBF

Trigger Group	VBF
period B-	Non-L1Topo photon+VBF HLT_g25_medium_L1EM22VHI_2j35_0eta490_bmv2c1077_split_2j35_0eta490 HLT_g25_medium_L1EM22VHI_j35_0eta490_bmv2c1077_split_3j35_0eta490_invm700 HLT_g25_medium_L1EM22VHI_4j35_0eta490_invm1000 L1Topo photon+VBF HLT_g20_tight_icaloose_j35_bmv2c1077_split_3j35_0eta490_invm500 (L1_EM18VHI_MJJ-300) Non-L1Topo b+VBF HLT_j55_gsc80_bmv2c1070_split_j45_gsc60_bmv2c1085_split_j45_320eta490 HLT_j45_gsc55_bmv2c1070_split_2j45_320eta490_L1J25.0ETA23_2J15.31ETA49 HLT_j80_0eta240_j60_j45_320eta490_AND_2j35_gsc45_bmv2c1070_split L1Topo b+VBF HLT_ht300_2j40_0eta490_invm700_L1HT150-J20s5.ETA31_MJJ-400-CF_AND_2j35_gsc45_bmv2c1070_
period K-	VBF inclusive HLT_j70_j50_0eta490_invm1100j70_dphi20_deta40_L1MJJ-500-NFF H->inv HLT_j70_j50_0eta490_invm1000j50_dphi24_xe90_pufit_xe50_L1MJJ-500-NFF H->tautau HLT_2e5_lhmedium_nod0_j70_j50_0eta490_invm900j50_L1MJJ-500-NFF HLT_2mu6_2j50_0eta490_invm900j50_L1MJJ-500-NFF HLT_e10_lhmedium_nod0_ivarloose_j70_j50_0eta490_invm900j50_L1MJJ-500-NFF HLT_mu10_ivarmed_j70_j50_0eta490_invm900j50_L1MJJ-500-NFF HLT_tau25_mediumRNN_tracktwoMVA_tau20_mediumRNN_tracktwoMVA_j70_j50_0eta490_invm900j50 H->bb HLT_j70_0eta490_j50_0eta490_2j35_0eta490_invm1000j50_L1MJJ-500-NFF_AND_2j35_bmv2c1070_split

Trigger Group	VBF
	EWK VBF SUSY HLT_e5_lhvloose_j70_j50_0eta490_invm1000j50_xe50_pufit_L1MJJ-500-NFF HLT_e5_lhvloose_nod0_j70_j50_0eta490_invm1000j50_xe50_pufit_L1MJJ-500-NFF* HLT_mu4_j70_j50_0eta490_invm1000j50_xe50_pufit_L1MJJ-500-NFF VBF+g (HBSM H->aa->ggjj, SM H(->bb)+g, SUSY GMSB VBF+g+MET) HLT_g35_medium_j70_j50_0eta490_invm900j50_L1MJJ-500-NFF

* Since run 358516 (period L)

Dedicated physics triggers

Trigger Group	Standard Model
Period B-	Wgg HLT_e24_lhmedium_nod0_2g12_loose HLT_2g10_loose_mu20

Trigger Group	Exotics
Period B-	LLP 3mu HLT_3mu6_msonly (L1_3MU6) primary HLT_3mu6_msonly_L1MU4_UNPAIRED_ISO support LLP photon+2mu HLT_g15_loose_2mu10_msonly (L1_2MU10) primary HLT_g15_loose_2mu10_msonly_L1MU4_UNPAIRED_ISO support LLP cal-ratio HLT_j30_jes_cleanLLP_PS_llp_L1TAU100 primary HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU100 primary HLT_j30_jes_cleanLLP_PS_llp_L1LLP-RO primary HLT_j30_jes_cleanLLP_PS_llp_noiso_L1LLP-RO primary HLT_j30_jes_cleanLLP_PS_llp_L1LLP-NOMATCH primary HLT_j30_jes_cleanLLP_PS_llp_noiso_L1LLP-NOMATCH primary HLT_j30_jes_cleanLLP_PS_llp_L1TAU30_EMPTY support HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU30_EMPTY support HLT_j30_jes_cleanLLP_PS_llp_L1TAU30_UNPAIRED_ISO support HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU30_UNPAIRED_ISO support LLP muvtx HLT_j30_muvtx (L1_2MU10) primary HLT_j30_muvtx_noiso (L1_2MU10) support LLP lepton+jets HLT_mu6_dR11_mu20_msonly_iloosems_mu6noL1_dR11_msonly (L1_MU11_2MU6) HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan05_L1MU20_J40 HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan05_L1MU20_XE30 HLT_mu20_msonly_iloosems_mu6noL1_msonly_nscan05_L1MU4_UNPAIRED_ISO Low mass leptophobic Z' in association with an ISR photon HLT_g85_tight_L1EM22VHI_3j50noL1

Trigger Group	Exotics
	HNL HLT_e9_lhvloose_nod0_mu20_mu8noL1* mono-pole HLT_g0_hiptrt_L1EM22VHI TLA HLT_j0_perf_ds1_L1J100 Z' HLT_mu80_msonly_3layersEC Late muons HLT_mu10_mgonly_L1LATE-MU10_J50 HLT_mu10_mgonly_L1LATE-MU10_XE40

* Chain name used since run 350067, HLT_mu20_mu8noL1_e9_lhvloose_nod0 used before

Trigger Group	Higgs
Period B- HLT_tau35_medium1_tracktwoEF_tau25_medium1_tracktwoEF_L1TAU20IM_2TAU12IM_4J12.0ETA23 g+tau for excl Higgs decays HLT_g25_medium_L1EM24VHI_tau25_dikaonmass_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion1loose_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion2_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi1_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi2_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_singlepion_tracktwo_50mVis10000 HLT_g35_medium_L1EM24VHI_tau25_dipion3_tracktwo_60mVis10000 e/mu+jets for VH HLT_e20_lhtight_nod0_ivarloose_3j20_L1EM18VHI_3J20 HLT_mu14_ivarloose_3j20_L1MU10_3J20 HLT_mu20_ivarmedium_L1MU10_2J15_J20 H->aa->bbmumu HLT_mu20_ivarmedium_mu8noL1 Low-mass diphoton HLT_2g6_tight_icalotight_L1J100 H->eeg HLT_e25_mergedtight_g35_medium_Heg	
period K- HLT_tau35_mediumRNN_tracktwoMVA_tau25_mediumRNN_tracktwoMVA_L1TAU20IM_2TAU12IM_4.	X->HH->tautau g+tau for excl Higgs decays HLT_g25_medium_L1EM24VHI_tau25_dikaonmass_tracktwoEF_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion1loose_tracktwoEF_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion2_tracktwoEF_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi1_tracktwoEF_50mVis10000

Trigger Group	Higgs
	HLT_g25_medium_L1EM24VHI_tau25_kaonpi2_tracktwoEF_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_singlepion_tracktwoEF_50mVis10000 HLT_g35_medium_L1EM24VHI_tau25_dipion3_tracktwoEF_60mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dikaonmass_tracktwoMVA_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion1loose_tracktwoMVA_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion2_tracktwoMVA_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi1_tracktwoMVA_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi2_tracktwoMVA_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_singlepion_tracktwoMVA_50mVis10000 HLT_g35_medium_L1EM24VHI_tau25_dipion3_tracktwoMVA_60mVis10000

Trigger Group	SUSY
Period B-	3b+MET HLT_j85_gsc100_bmv2c1050_split_xe85_pufit_xe50_L1XE55 HLT_2j35_gsc45_bmv2c1070_split_xe85_pufit_xe50_L12J15_XE55 HLT_3j35_bmv2c1077_split_xe70_pufit_xe50_L13J15.0ETA25_XE40 Higgsino single e HLT_e5_lhloose_nod0_j50_xe70_pufit_2dphi10_L1J40_XE50_DPFI-J20s2XE50 HLT_e5_lhloose_nod0_j40_xe70_pufit_2dphi10_L1XE60 (non-L1Topo backup) single mu HLT_mu4_j90_xe90_pufit_2dphi10_L1MU4_J50_XE50_DPFI-J20s2XE30 HLT_mu4_j90_xe90_pufit_2dphi10_L1MU4_XE60 (non-L1Topo backup) di-e HLT_2e5_lhvloose_nod0_j40_xe70_pufit_2dphi10_L1J40_XE50_DPFI-J20s2XE50 HLT_2e5_lhvloose_nod0_j40_xe70_pufit_2dphi10_L1XE60 (non-L1Topo backup) di-mu HLT_2mu4_invm1_j20_xe40_pufit_2dphi10_L12MU4_J20_XE30_DPFI-J20s2XE30 HLT_2mu4_invm1_j20_xe60_pufit_2dphi10_L12MU4_J40_XE50 (non-L1Topo backup) emu HLT_e5_lhvloose_nod0_mu4_j30_xe40_pufit_2dphi10_L1MU4_J30_XE40_DPFI-J20s2XE30 HLT_e5_lhmedium_nod0_mu4_j30_xe65_pufit_2dphi10_L1MU4_XE60 (non-L1Topo backup) stopped gluinos HLT_j55_0eta240_xe50_L1J30_EMPTY HLT_j55_0eta240_xe50_L1J30_FIRSTEMPTY photon+multijet HLT_g45_loose_6j45_0eta240

Trigger Group	Egamma
Period B-	Z T&P HLT_e26_lhtight_nod0_e15_etcut_L1EM7_Zee

Triggers in 2017

Introduction

- History of menu evolution is documented in Recommendation for analysis groups. Important changes are;
 - ◆ Enabled the HLT full menu from Period B1- (Run325713 with 600b runs on Mon Jun 05 2017)
 - ◆ Enabled L1Topo triggers for physics from Period B5-
 - ◆ Switched to the 1.7e34 primaries from Period B8-
-  **IMPORTANT:** Triggers that will become the lowest un-prescaled with the increase of luminosity are also included for lower luminosities. To check which will be the lowest un-prescaled as a function of luminosity, please refer to this twiki Menu Evolution.
-  **IMPORTANT:** as of 2017 November, a python interface to obtain information about the lowest-unprescaled triggers for a given signature and a given period is available. See the twiki for the instruction.
- Known issues are detailed in TriggerIssues2017.

Data Periods

Some useful COMA ConditionsMetadata links for data17_13TeV are:

- COMA Periods of data17_13TeV [↗](#)
- COMA Period Runs in data17_13TeV [↗](#)
- COMA Period Triggers in data17_13TeV [↗](#)
- COMA Period Streams in data17_13TeV [↗](#)

Rulebook tags

We tag the rulebook package in order to be able to trace chain rules when a new AthenaP1 release is deployed and used online. Please, find below the tags corresponding to different releases and periods together with the corresponding JIRAs.

- Tag for AthenaP1, 21.1.3 (period A1) : May Startup up to 12b [↗](#) (3b ATR-16365 [↗](#), 12b ATR-16384 [↗](#))
- Tag for AthenaP1, 21.1.4 (period A2-B1) : June Startup up to 600b [↗](#) (72b magnets-off runs ATR-16418 [↗](#), 300b toroid-off runs ATR-16432 [↗](#), 600b with HLT enabled ATR-16454 [↗](#))
- Tag for AthenaP1, 21.1.5 (period B2-B4) : June afterScrubbing up to 900b [↗](#) (9b, 600b, 900b, ATR-16500 [↗](#))
- Tag for AthenaP1, 21.1.6 (period B4-B8) : June Rampup up to 1900b [↗](#), 2017June_Rampup up to 2550b [↗](#) (1200b, 1500b, ATR-16530 [↗](#), 1700b-2550b ATR-16552 [↗](#))
- Tag for AthenaP1, 21.1.8 (period C1-C8) : July after TS1 up to 600b [↗](#) (high-mu up to 600b ATR-16683 [↗](#)), July 600b low-mu [↗](#) (low-mu 600b ATR-16399 [↗](#)), July post-TS1 2550b [↗](#) (post-TS1 2550b ATR-16695 [↗](#))
- Tag for AthenaP1, 21.1.9 (period D2) : July post-MD2 VdM scan [↗](#) (ATR-16757 [↗](#)), post-MD2 600b high-mu [↗](#) (ATR-16778 [↗](#)), post-MD2 600b low-mu mu(ave)~1 [↗](#) (ATR-16772 [↗](#))
- Tag for AthenaP1, 21.1.8 (period D3-D6) : August post-MD2 2550b [↗](#) (ATR-16787 [↗](#)), Toroid-off 2550b disabling L1_MU [↗](#) (ATR-16808 [↗](#))
- Tag for AthenaP1, 21.1.10- (period X) : August various numbers of bunches [↗](#) (ATR-16842 [↗](#))
- Tag for AthenaP1, 21.1.19 : Final RB in 2017 high-mu run on 8b4e [↗](#) (ATR-17370 [↗](#)), LowMu/5TeV/ALFA runs in Nov [↗](#) (ATR-17208, 5TeV run [↗](#), ATR-17253, 13TeV mu=2,1 run [↗](#), ATR-17210, ALFA run [↗](#))

Egamma

Trigger Group	Single electron	Multi electron
period B-	HLT_e26_lhtight_nod0_ivarloose HLT_e60_lhmedium_nod0 HLT_e140_lhloose_nod0 HLT_e300_etcut	HLT_2e17_lhvloose_nod0_L12EM15VHI* HLT_2e24_lhvloose_nod0 HLT_e24_lhvloose_nod0_2e12_lhvloose_nod0_L1EM20VH_3EM10VH

Trigger Group	Single photon	Multi photon
period B-	HLT_g140_loose HLT_g300_etcut	HLT_2g20_tight_icalovloose_L12EM15VHI* HLT_2g22_tight_L12EM15VHI* HLT_2g25_tight_L12EM20VH HLT_g35_medium_g25_medium_L12EM20VH HLT_2g50_loose_L12EM20VH HLT_2g25_loose_g15_loose

* L1_2EM15VHI was accidentally prescaled in periods B5-B8 (runs 326834-328393 with an effective reduction of 0.6 fb⁻¹)

Egamma + MET/Jets

Trigger Group	electron+MET	photon+MET
period B-	HLT_e70_lhloose_nod0_xe70noL1	HLT_g80_loose_xe80noL1

Trigger Group	photon+Jets
period B-	HLT_g85_tight_L1EM22VHI_3j50noL1

Muon

Trigger Group	Single muon	Multi muon
period B-	HLT_mu26_ivarmedium HLT_mu50 HLT_mu60_0eta105_msonly	HLT_2mu14 HLT_mu22_mu8noL1 HLT_mu22_mu8noL1_calotag_0eta010 HLT_mu20_2mu4noL1 HLT_3mu6 HLT_3mu6_msonly HLT_4mu4

Electron + Muon

Trigger	e+mu	e+2mu	2e+mu
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Group			
period	HLT_e17_lhloose_nod0_mu14	HLT_e12_lhloose_nod0_2mu10	HLT_2e12_lhloose_nod0_mu10
B-	HLT_e26_lhmedium_nod0_mu8noL1		
	HLT_e7_lhmedium_nod0_mu24		

Electron/Muon + Photon

Trigger Group	electron+photon	muon+photon
period	HLT_e24_lhmedium_nod0_L1EM20VH_g25_medium	HLT_g25_medium_mu24
B-	HLT_e25_mergedtight_g35_medium_Heg	HLT_g15_loose_2mu10_msonly
		HLT_g35_loose_L1EM24VHI_mu18
		HLT_g35_loose_L1EM24VHI_mu15_mu2noL1
		HLT_g35_tight_icalotight_L1EM24VHI_mu18noL1
		HLT_g35_tight_icalotight_L1EM24VHI_mu15noL1

Bphysics

Complete table of Bphys triggers including prescaled triggers together with luminosity per trigger per period can be found at this link

Trigger Group	Generic	B->mumu
period	HLT_mu11_mu6_bDimu	HLT_mu11_mu6_bBmumu
B-	HLT_mu11_mu6_bDimu2700	HLT_2mu6_bBmumu_Lxy0_L1BPH-2M9-2MU6_BPH-2DR15-2MU6
	HLT_mu11_mu6_bDimu2700_Lxy0	

* Included in Main and not in BphysLS stream

Taus

Trigger Group	Single tau	di-tau+L1 jets
period	HLT_tau160_medium1_tracktwo	HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo
B1-B4		
period		HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo_0
B5-B7		HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo_L
		HLT_tau40_medium1_tracktwo_tau35_medium1_tracktwo
period	HLT_tau160_medium1_tracktwo_L1TAU100	
B8-		

HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo ran unprescaled all the time when the peak lumi of the run was below 1.2e34.

Trigger Group	tau+muon
period B	HLT_mu14_ivarloose_tau25_medium1_tracktwo (L1_MU10_TAU12IM_J25_2J12)
	HLT_mu14_ivarloose_tau35_medium1_tracktwo (L1_MU10_TAU20IM)
	HLT_mu14_ivarloose_tau25_medium1_tracktwo_L1MU10_TAU12IM_3J12

Trigger Group	tau+MET	tau+electron+MET
period B1-B7	HLT_tau35_medium1_tracktwo_xe70_L1XE45 HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo_xe50	HLT_e17_lhmedium_nod0_tau25_medium1_tracktwo_xe50
period B8-B15	HLT_tau35_medium1_tracktwo_xe70_L1XE45 HLT_tau60_medium1_tracktwo_tau25_medium1_tracktwo_xe50	

Trigger Group	Single jet	Multi jets	
period B (all runs except 325789, 327761, 328017)	small-R HLT_j380 (+) HLT_j400 HLT_j225_gsc380_boffperf_split (+) HLT_j225_gsc400_boffperf_split large-R trimmed HLT_j420_a10t_lcw_jes_L1J100 (+) (++) HLT_j440_a10t_lcw_jes_L1J100 (++) HLT_j390_a10t_lcw_jes_30smcINF_L1J100 (++) large-R untrimmed HLT_j420_a10_lcw_subjes_L1J100 (+) HLT_j440_a10_lcw_subjes_L1J100 large-R reclustered HLT_j420_a10r_L1J100 (+) HLT_j440_a10r_L1J100	small-R HLT_3j200 HLT_4j100 (+) 4j120 HLT_4j60_gsc100_boffperf_split (+) HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) HLT_5j50_gsc70_boffperf_split_L14J150ETA25 HLT_6j60_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) HLT_6j45_gsc60_boffperf_split_L14J150ETA25 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA25 HLT_10j40_L14J15 small-R central HLT_5j65_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20)	HLT_3j200 HLT_4j100 (+) 4j120 HLT_4j60_gsc100_boffperf_split (+) HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) HLT_5j50_gsc70_boffperf_split_L14J150ETA25 HLT_6j60_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) HLT_6j45_gsc60_boffperf_split_L14J150ETA25 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA25 HLT_10j40_L14J15 HLT_5j65_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20)

Trigger Group	Single jet	Multi jets	
		HLT_5j50_gsc65_boffperf_split_0eta240_L14J150ETA25 HLT_6j45_0eta240 (L14J15) (+) HLT_6j45_0eta240_L14J20) (++) HLT_6j50_0eta240_L14J15 HLT_6j25_gsc45_boffperf_split_0eta240_L14J20 (+) HLT_6j25_gsc45_boffperf_split_0eta240_L14J150ETA25 (+) HLT_6j25_gsc50_boffperf_split_0eta240_L14J20 HLT_6j25_gsc50_boffperf_split_0eta240_L14J150ETA25 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (++)	
period B (all runs except 325789)	small-R HLT_j420 HLT_j225_gsc420_boffperf_split large-R trimmed HLT_j460_a10t_lcw_jes_L1J100 HLT_j420_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j460_a10_lcw_subjes_L1J100 large-R reclustered HLT_j460_a10r_L1J100	small-R HLT_3j200 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j85_L14J15 HLT_5j60_gsc85_boffperf_split (L14J20) HLT_5j60_gsc85_boffperf_split_L14J150ETA25 HLT_6j70_L14J15 HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA25 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA25 HLT_10j40_L14J15 small-R central HLT_5j70_0eta240_L14J15 HLT_5j50_gsc70_boffperf_split_0eta240 (L14J20) HLT_5j50_gsc70_boffperf_split_0eta240_L14J150ETA25 HLT_6j55_0eta240_L14J15 HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L15J150ETA25 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (++)	HT (* HT (*
period B (all runs)	small-R HLT_j450 HLT_j225_gsc440_boffperf_split large-R trimmed HLT_j480_a10t_lcw_jes_L1J100 HLT_j440_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j480_a10_lcw_subjes_L1J100 large-R reclustered HLT_j480_a10r_L1J100	small-R HLT_3j225 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j90 (L14J20) HLT_5j90_L14J150ETA25 HLT_5j60_gsc90_boffperf_split (L14J20) HLT_5j60_gsc90_boffperf_split_L14J150ETA25 HLT_6j70 (L14J20) HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA25 HLT_7j45 (L16J15) HLT_7j45_L14J150ETA25 HLT_7j45_L14J20 HLT_7j25_gsc45_boffperf_split_L14J20	HT (* HT (*

Trigger Group	Single jet	Multi jets	
		HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L14J150ETA25 (++) HLT_7j25_gsc45_boffperf_split_L14J150ETA25 large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (++) HLT_2j330_a10t_lcw_jes_40smcINF_L1J100	
period D1-D5	small-R HLT_j400 (++) HLT_j420 HLT_j225_gsc400_boffperf_split (++) HLT_j225_gsc420_boffperf_split large-R trimmed HLT_j440_a10t_lcw_jes_L1J100 (++) HLT_j460_a10t_lcw_jes_L1J100 HLT_j390_a10t_lcw_jes_30smcINF_L1J100 (++) HLT_j420_a10t_lcw_jes_40smcINF_L1J100 large-R untrimmed HLT_j440_a10_lcw_subjes_L1J100 (++) HLT_j460_a10_lcw_subjes_L1J100 large-R reclustered HLT_j440_a10r_L1J100 (++) HLT_j460_a10r_L1J100	small-R HLT_3j200 HLT_4j120 HLT_4j60_gsc115_boffperf_split HLT_5j70_L14J15 (++) HLT_5j85_L14J15 HLT_5j50_gsc70_boffperf_split (L14J20) (++) HLT_5j50_gsc70_boffperf_split_L14J150ETA25 (++) HLT_5j60_gsc85_boffperf_split (L14J20) HLT_5j60_gsc85_boffperf_split_L14J150ETA25 HLT_6j60_L14J15 (++) HLT_6j70_L14J15 HLT_6j45_gsc60_boffperf_split (L14J20) (++) HLT_6j45_gsc60_boffperf_split_L14J150ETA25 (++) HLT_6j50_gsc70_boffperf_split (L14J20) HLT_6j50_gsc70_boffperf_split_L14J150ETA25 HLT_7j45_L14J15 HLT_7j25_gsc45_boffperf_split_L14J20 HLT_7j25_gsc45_boffperf_split_L14J150ETA25 HLT_10j40_L14J15 small-R central HLT_5j65_0eta240_L14J15 (++) HLT_5j70_0eta240_L14J15 HLT_5j50_gsc65_boffperf_split_0eta240 (L14J20) (++) HLT_5j50_gsc65_boffperf_split_0eta240_L14J150ETA25 (++) HLT_5j50_gsc70_boffperf_split_0eta240 (L14J20) HLT_5j50_gsc70_boffperf_split_0eta240_L14J150ETA25 HLT_6j50_0eta240_L14J15 (++) HLT_6j55_0eta240_L14J15 HLT_6j25_gsc50_boffperf_split_0eta240_L14J20 (++) HLT_6j25_gsc50_boffperf_split_0eta240_L14J150ETA25 (++) HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L14J150ETA25 (++) large-R trimmed HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (++) HLT_2j330_a10t_lcw_jes_40smcINF_L1J100	HLT_6j35_gsc55_boffperf_split_0eta240_L14J20 HLT_6j35_gsc55_boffperf_split_0eta240_L14J150ETA25 (++) HLT_7j25_gsc45_boffperf_split_L14J150ETA25 HLT_2j330_a10t_lcw_jes_30smcINF_L1J100 (++) HLT_2j330_a10t_lcw_jes_40smcINF_L1J100
period D6-K			

- (*) MET trigger option that gives you the best combination of low thresholds & small loss of luminosity (~270 pb⁻¹ loss of luminosity w.r.t. option ***)
- (**) MET trigger option that gives you lower thresholds (L1_XE50) but has been prescaled for part of the runs (~ 900 pb⁻¹ loss of luminosity w.r.t. option ***)
- (***) MET trigger option that gives you full luminosity at the cost of a higher threshold (recommended if you prefer not to combine triggers over different data-taking periods)
- (+) Jet trigger also disabled in runs 327582, 327745, 327860, 328221 and 328374 of period B (~1.2ifb loss)
- (++) Jet trigger slightly different combination of missing runs / part-enabling of trigger for different parts of some runs
- (+++) Jet trigger missing small fraction of luminosity from periods C (0.14 from 2.77 ifb) and D1-D5 (0.35 from 5.90 ifb)

Important remarks:

- The above quoted luminosity numbers are only indicative. Always assess the luminosity yourself to weigh threshold vs. luminosity arguments for your analysis.
- Further to this, please see JetTriggersToUseIn2017 coupled with the 1.7 and 1.5e34 menus from L34TriggerMenu2017 for the jet trigger group's recommendations and the available options.
- Before run 332655 (the deployment of release 21.1.10) jet GSC triggers were off for a few lumi blocks just after the start of the run while the beamspot determination was under way. A special GRL is provided for analysers using these triggers, removing 262 of the 88487 lumi blocks, corresponding to 73 pb⁻¹ out of the total 40404.4 pb⁻¹ in the data17_13TeV.periodAllYear_DetStatus-v97-pro21-13_Unknown_PHYS_StandardGRL_All_Good_25ns.xml GRL
- Analyses using MET triggers should also check events that fire **HLT_noalg_L1J400** to see if any of these events were **not** fired by the met trigger in the analysis phase-space.

Bjet

Trigger Group	Single bjet	Multi bjet
period B2-B7	(1b) HLT_j175_gsc225_bmv2c1040_split HLT_j225_gsc275_bmv2c1070_split HLT_j225_gsc300_bmv2c1077_split HLT_j225_gsc360_bmv2c1085_split (1b+3j) HLT_j50_gsc65_bmv2c1040_split_3j50_gsc65_boffperf_split HLT_j60_gsc85_bmv2c1050_split_3j60_gsc85_boffperf_split	(2b) HLT_j150_gsc175_bmv2c1070_split_j45 (2b+1j) HLT_j110_gsc150_boffperf_split_2j35_gsc55 (2b+2j) HLT_2j15_gsc35_bmv2c1050_split_2j15_gsc35 HLT_2j25_gsc45_bmv2c1060_split_2j25_gsc45 HLT_2j35_gsc55_bmv2c1070_split_2j35_gsc55 (2b+3j) HLT_2j15_gsc35_bmv2c1060_split_3j15_gsc35 HLT_2j25_gsc45_bmv2c1070_split_3j25_gsc45 (3b+1j) HLT_3j15_gsc35_bmv2c1070_split_j15_gsc35 (4b) HLT_4j15_gsc35_bmv2c1077_split_L14 HLT_2j15_gsc35_bmv2c1070_split_2j15_gsc35 (3b) HLT_3j50_gsc65_bmv2c1077_split_L13
period B8-	(1b) HLT_j175_gsc225_bmv2c1040_split HLT_j225_gsc275_bmv2c1060_split HLT_j225_gsc300_bmv2c1070_split HLT_j225_gsc360_bmv2c1077_split (1b+3j) HLT_j55_gsc75_bmv2c1040_split_3j55_gsc75_boffperf_split	(2b) HLT_j150_gsc175_bmv2c1060_split_j45 (2b+1j) HLT_j110_gsc150_boffperf_split_2j35_gsc55 (2b+2j) HLT_2j15_gsc35_bmv2c1040_split_2j15_gsc35 (2b+3j) HLT_2j15_gsc35_bmv2c1050_split_3j15_gsc35

Trigger Group	Single bjet	Multi bjet
	HLT_j60_gsc85_bmv2c1050_split_3j60_gsc85_boffperf_split	HLT_2j25_gsc45_bmv2c1060_split_3j25_gsc45_boffperf_split HLT_2j25_gsc45_bmv2c1050_split_2j25_gsc45_boffperf_split HLT_2j35_gsc55_bmv2c1060_split_2j35_gsc55_boffperf_split (3b+1j) HLT_3j15_gsc35_bmv2c1070_split_j15_gsc35_boffperf_split (4b) HLT_4j15_gsc35_bmv2c1077_split_L14 HLT_2j15_gsc35_bmv2c1070_split_2j15_gsc35_boffperf_split (3b) HLT_3j50_gsc65_bmv2c1077_split_L13

* Updated the b-jet triggers with MV2c10 from Period B2-

Bjet + HT/MET

Trigger Group	Bjet+HT	Bjet + MET
period B2-B4	-	HLT_j60_gsc100_bmv2c1050_split_xe8 HLT_2j25_gsc45_bmv2c1070_split_xe8 HLT_3j15_gsc35_bmv2c1077_split_xe8
period B5-	HLT_j35_gsc55_bmv2c1050_split_ht700_L1HT190-J15s5.ETA21 HLT_2j35_gsc55_bmv2c1050_split_ht300_L1HT190-J15s5.ETA21 HLT_2j35_gsc55_bmv2c1060_split_ht300_L1HT190-J15s5.ETA21 (below 1.5e34)	

VBF

Trigger Group	photon+VBF (non-L1Topo)	
period B1-B4	HLT_g25_medium_L1EM22VHI_2j35_0eta490_bmv2c1077_split_2j35_0eta490 HLT_g25_medium_L1EM22VHI_j35_0eta490_bmv2c1077_split_3j35_0eta490_invm700 HLT_g25_medium_L1EM22VHI_4j35_0eta490_invm1000	-
period B5-B8		-
period C-		HLT_g20_tight_i

Dedicated physics triggers

Standard Model

Trigger Group	Wgg
Period B-	HLT_e24_1hmedium_nod0_2g12_loose HLT_2g10_loose_mu20

Exotics

Trigger Group	LLP cal-ratio (non-L1Topo)	LLP cal-ratio (L1Topo)
Period B1-B7	HLT_j30_jes_cleanLLP_PS_llp_L1TAU60	-
	HLT_j30_jes_cleanLLP_PS_llp_pufix_L1TAU60	-
	HLT_j30_jes_cleanLLP_PS_llp_pufix_noiso_L1TAU60	-

Trigger Group	LLP cal-ratio (non-L1Topo)	LLP cal-ratio (L1Topo)
	HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU60 HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU60	
Period B8-	HLT_j30_jes_cleanLLP_PS_llp_L1TAU100 HLT_j30_jes_cleanLLP_PS_llp_puifix_L1TAU100 HLT_j30_jes_cleanLLP_PS_llp_puifix_noiso_L1TAU100 HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU100 HLT_j30_jes_cleanLLP_PS_llp_noiso_L1TAU100	HLT_j30_jes_cleanLLP_PS_llp_L1LLP-RO HLT_j30_jes_cleanLLP_PS_llp_noiso_L1LLP-RO HLT_j30_jes_cleanLLP_PS_llp_L1LLP-NOMAT HLT_j30_jes_cleanLLP_PS_llp_noiso_L1LLP-NOMAT

Trigger Group	lepton+jets	Late muons
Period B	HLT_mu20_msonly_mu15noL1_msonly_nscan05_noComb (below 1.0e34) HLT_mu20_msonly_ilooseems_mu6noL1_msonly_nscan05_L1MU20_J40 HLT_mu20_msonly_ilooseems_mu6noL1_msonly_nscan05_L1MU20_XE30	-
Period C-	HLT_mu6_dR11_mu20_msonly_ilooseems_mu6noL1_dR11_msonly (seed fix since 21.1.10) HLT_mu20_msonly_mu15noL1_msonly_nscan05_noComb (below 1.0e34) HLT_mu20_msonly_ilooseems_mu6noL1_msonly_nscan05_L1MU20_J40 HLT_mu20_msonly_ilooseems_mu6noL1_msonly_nscan05_L1MU20_XE30	HLT_mu10_mgonly_L1LATE- HLT_mu10_mgonly_L1LATE-

Higgs

Trigger Group	g+tau for excl Higgs decays	e/mu+j
Period B-	HLT_g25_medium_L1EM24VHI_tau25_dikaonmass_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion1loose_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_dipion2_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi1_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_kaonpi2_tracktwo_50mVis10000 HLT_g25_medium_L1EM24VHI_tau25_singlepion_tracktwo_50mVis10000 HLT_g35_medium_L1EM24VHI_tau25_dipion3_tracktwo_60mVis10000	HLT_e20_lhtight_nod0_ivar HLT_mu14_ivarloose_3j20_ HLT_mu20_ivarmedium_L1

SUSY

Trigger Group	bjets+MET	
Period B1-B4	HLT_j60_gsc100_bmv2c1050_split_xe80_mht_L1XE60 HLT_2j25_gsc45_bmv2c1070_split_xe80_mht_L12J15_XE55 HLT_3j15_gsc35_bmv2c1077_split_xe60_mht_L13J15.0ETA25_XE40	HLT_2mu4_invm1_j20_xe60_pufit_ HLT_mu4_j90_xe90_pufit_2dphi10_ HLT_e5_lhmedium_nod0_mu4_j30_ HLT_2e5_lhvloose_nod0_j40_xe70_ HLT_e5_lhloose_nod0_j40_xe70_p
Period B5-		HLT_2mu4_invm1_j20_xe40_pufit_ HLT_mu4_j90_xe90_pufit_2dphi10_ HLT_e5_lhvloose_nod0_mu4_j30_x HLT_2e5_lhvloose_nod0_j40_xe70_ HLT_e5_lhloose_nod0_j50_xe70_p HLT_2mu4_invm1_j20_xe60_pufit_ HLT_mu4_j90_xe90_pufit_2dphi10_ HLT_e5_lhmedium_nod0_mu4_j30_ HLT_2e5_lhvloose_nod0_j40_xe70_ HLT_e5_lhloose_nod0_j40_xe70_p

Egamma

Trigger	Z T&P	W T&P
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Group		
Period B-	HLT_e26_lhtight_nod0_e15_etcut_L1EM7_Zee	-
21.1.10-		HLT_e60_etcut_trkcut_L1EM24VHIM_j15_perf_xe60_6d HLT_e60_etcut_trkcut_L1EM24VHIM_xs30_j15_perf_xe

Triggers in 2016

Introduction

- **Note:** Lowest unprescaled **combined** and **special triggers** can be found in the pdf table attached at the bottom of the Twiki page on pages 8, 9 & 10 (rather than in the tables below).
-  **IMPORTANT:** Triggers that will become the lowest un-prescaled with the increase of luminosity are also included for lower luminosities. To check which will be the lowest un-prescaled as a function of luminosity, please refer to this twiki Menu Evolution.

Data Periods

Some useful COMA ConditionsMetadata links for data16_13TeV are:

- COMA Periods of data16_13TeV [↗](#)
- COMA Period Runs in data16_13TeV [↗](#)
- COMA Period Triggers in data16_13TeV [↗](#)
- COMA Period Streams in data16_13TeV [↗](#)

Some important additional Period-specific notes:

-

Rulebook tags

We started to tag the rulebook package in order to be able to trace chain rules when a new AtlasP1HLT cache is deployed and used online. Please, find below the tags corresponding to different releases and periods together with the corresponding JIRAs.

- Tag for AtlasP1HLT, 20.11.0.9 (period A1-A2) : TrigMenuRulebook-00-05-38 [↗](#) ATR-13864 [↗](#)
ATR-13894 [↗](#) ATR-13874 [↗](#)
- Tag for AtlasP1HLT, 20.11.0.9 (period A3) : TrigMenuRulebook-00-05-44 [↗](#) ATR-13947 [↗](#)

Egamma (+MET)

Trigger Group	Single electron	Multi electron	S
period A	HLT_e24_lhtight_nod0_ivarloose* HLT_e60_lhmedium_nod0 HLT_e60_medium HLT_e140_lhloose_nod0	HLT_2e15_lhvloose_nod0_L12EM13VH** HLT_e17_lhloose_nod0_2e9_lhloose_nod0	HL HL

Trigger Group	Single electron	Multi electron	
	HLT_e300_etcut		
period B-D3	HLT_e24_lhtight_nod0_ivarloose* HLT_e60_lhmedium_nod0 HLT_e60_medium HLT_e140_lhloose_nod0 HLT_e300_etcut	HLT_2e15_lhvloose_nod0_L12EM13VH** HLT_e17_lhloose_nod0_2e9_lhloose_nod0	HL HL
period D4-F	HLT_e26_lhtight_nod0_ivarloose HLT_e60_lhmedium_nod0 HLT_e60_medium HLT_e140_lhloose_nod0 HLT_e300_etcut	HLT_2e17_lhvloose_nod0 HLT_e17_lhloose_nod0_2e9_lhloose_nod0	HL HL
period G-	HLT_e26_lhtight_nod0_ivarloose HLT_e60_lhmedium_nod0 HLT_e60_medium HLT_e140_lhloose_nod0 HLT_e300_etcut	HLT_2e17_lhvloose_nod0 HLT_e17_lhloose_nod0_2e9_lhloose_nod0*** HLT_e17_lhloose_nod0_2e10_lhloose_nod0_L1EM15VH_3EM8VH	HL HL

* In run 298687 L1_EM20VHI was partially prescaled (due to an autoprescaler problem). To get the full statistics for this run an OR of HLT_e24_lhmedium_nod0_L1EM20VH, HLT_e24_lhtight_nod0_ivarloose is required.

** In run 300540 L1_2EM13VH used in HLT_2e15_lhvloose_nod0_L12EM13VH was prescaled to 10 in LB 210-260 due to high dead-times related to pixel readout. HLT_2e17_lhvloose_nod0 (seeded by L1_2EM15VH) can be used instead to avoid any prescales.

*** Triggers disabled when lumi>1.2e34 could to apply a logical OR between:

HLT_e17_lhloose_nod0_2e9_lhloose_nod0 ||

HLT_e17_lhloose_nod0_2e10_lhloose_nod0_L1EM15VH_3EM8VH

HLT_e20_lhmedium_nod0_2g10_loose || HLT_e20_lhmedium_nod0_2g10_loose_L1EM15VH_3EM8VH to recover ~1fb-1 of missing data, BUT we are not able to provide scale-factors for the later triggers.

Muon

Trigger Group	Single muon	Multi muon
period A	HLT_mu24_iloose (called _L1MU15 in MC15c) HLT_mu24_ivarloose (called _L1MU15 in MC15c) HLT_mu40 HLT_mu50	HLT_2mu10 HLT_2mu10_nomucomb HLT_mu20_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu4 HLT_mu6_2mu4 HLT_mu20_nomucomb_mu6noL1_nscan03 HLT_mu11_nomucomb_2mu4noL1_nscan03_L1MU11_2MU6 HLT_mu20_msonly_mu10noL1_msonly_nscan05_noComb HLT_mu11_nomucomb_2mu4noL1_nscan03_L1MU11_2MU6_bTau HLT_mu11_nomucomb_mu6noL1_nscan03_L1MU11_2MU6_bTau HLT_mu6_nomucomb_2mu4_nomucomb_bTau_L1MU6_3MU4 HLT_2mu6_nomucomb_mu4_nomucomb_bTau_L12MU6_3MU4

Trigger Group	Single muon	Multi muon
period B-D3	HLT_mu24_ivarmedium HLT_mu24_imedium HLT_mu50	HLT_2mu14 HLT_2mu14_nomucomb HLT_mu20_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu6 HLT_mu6_2mu4 HLT_mu20_nomucomb_mu6noL1_nscan03 HLT_mu11_nomucomb_2mu4noL1_nscan03_L1MU11_2MU6 HLT_mu20_msonly_mu10noL1_msonly_nscan05_noComb HLT_mu11_nomucomb_2mu4noL1_nscan03_L1MU11_2MU6_bTau HLT_mu11_nomucomb_mu6noL1_nscan03_L1MU11_2MU6_bTau HLT_mu6_nomucomb_2mu4_nomucomb_bTau_L1MU6_3MU4 HLT_2mu6_nomucomb_mu4_nomucomb_bTau_L12MU6_3MU4
period D4-E	HLT_mu24_ivarmedium/imedium* HLT_mu26_ivarmedium/imedium HLT_mu50	HLT_2mu14 HLT_mu20_mu8noL1* HLT_mu22_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu6_msonly
period F-G2	HLT_mu26_ivarmedium/imedium HLT_mu50	HLT_2mu14 HLT_mu22_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu6_msonly
period G3-I3	HLT_mu26_ivarmedium HLT_mu50	HLT_2mu14 HLT_mu22_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu6_msonly
period I4-	HLT_mu26_ivarmedium HLT_mu50	HLT_2mu14 HLT_mu22_mu8noL1 HLT_mu20_2mu4noL1 HLT_3mu6_msonly HLT_3mu4

*Prescaled when lumi>1.0e34

Electron + Muon

Trigger Group	e+mu
period A	HLT_e17_lhloose_nod0_mu14 HLT_e24_lhmedium_nod0_L1EM20VHI_mu8noL1 HLT_e7_lhmedium_nod0_mu24 HLT_e12_lhloose_nod0_2mu10 HLT_2e12_lhloose_nod0_mu10
period B-D3	HLT_e17_lhloose_nod0_mu14 HLT_e24_lhmedium_nod0_L1EM20VHI_mu8noL1 HLT_e7_lhmedium_nod0_mu24 HLT_e12_lhloose_nod0_2mu10 HLT_2e12_lhloose_nod0_mu10
period D4-	HLT_e17_lhloose_nod0_mu14 HLT_e26_lhmedium_nod0_L1EM22VHI_mu8noL1 HLT_e7_lhmedium_nod0_mu24

Trigger Group	e+mu
	HLT_e12_lhloose_nod0_2mu10 HLT_2e12_lhloose_nod0_mu10

Muon + Photon

Trigger Group	muon+photon	multi-muon+multi-photon
period A	HLT_g25_medium_mu24	HLT_g15_loose_2mu10_msonly HLT_2g10_loose_mu20 HLT_g35_loose_L1EM22VHI_mu15noL1_mu2noL1 HLT_g35_loose_L1EM22VHI_mu18noL1
period B-D3	HLT_g25_medium_mu24	HLT_g15_loose_2mu10_msonly HLT_2g10_loose_mu20 HLT_g35_loose_L1EM22VHI_mu15noL1_mu2noL1 HLT_g35_loose_L1EM22VHI_mu18noL1
period D4-	HLT_g25_medium_mu24	HLT_g15_loose_2mu10_msonly HLT_2g10_loose_mu20 HLT_g35_loose_L1EM22VHI_mu15noL1_mu2noL1 HLT_g35_loose_L1EM22VHI_mu18noL1

Bphysics

Complete table of Bphys triggers including prescaled triggers together with luminosity per trigger per period can be found at this link

Trigger Group	Bmumu	Jpsimumu	Upsimumu
period A	HLT_mu6_mu4_bBmumu	HLT_mu6_mu4_bJpsimumu (Until A10 299584 (PS: 1--4))	HLT_2mu6_bUpsimumu
period B-D3	HLT_2mu6_bBmumu	HLT_mu10_mu6_bJpsimumu	HLT_mu10_mu6_bUpsimumu
period D4-	HLT_mu10_mu6_bBmumuxv2_noL2 (MU10_2MU6)	HLT_2mu10_bJpsimumu_noL2_L12MU10	

* Introduced BphysDelayed stream from period D

Taus (+ electron, muon, MET)

Trigger Group	Single tau	di-tau	
period A	HLT_tau80_medium1_tracktwo_L1TAU60	HLT_tau35_loose1_tracktwo_tau25_loose1_tracktwo (with J25)	HLT_tau35_loose1_tracktwo_tau25_loose1_tracktwo (with J25)
period B-D3	HLT_tau125_medium1_tracktwo	HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo (with J25)	HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo (with J25)

Trigger Group	Single tau	di-tau	
period D4-	HLT_tau160_medium1_tracktwo (tau125 almost unprescaled*)	HLT_tau80_medium1_TAU60_tau50_medium1_L1TAU12 HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo (TAU20IM_2TAU12IM_J25_2J20_3J12)	HL J25 HL

* run lumical with the correct triggers to get the right int lumi!

Jets, MET, Jet+MET, HT

Trigger Group	Single jet	Multi jets	MET	HT
period A	HLT_j340 HLT_j360_a10r_L1J100 HLT_j360_a10_lcw_L1J100 HLT_j260_320eta490	HLT_3j175 HLT_4j85 HLT_5j55 HLT_6j45 HLT_7j45 HLT_10j40_L14J20	HLT_xe90_mht_L1XE50 (a)	HLT_ht850_L1J7
period B-C	HLT_j380 HLT_j400_a10r_L1J100 HLT_j400_a10_lcw_L1J100 HLT_j260_320eta490	HLT_3j175 HLT_4j100 HLT_5j70_L14J15 HLT_6j60 HLT_6j45_0eta240 HLT_7j45 HLT_10j40_L14J20	HLT_xe90_mht_L1XE50	HLT_ht1000_L1J10
period D1-D3	HLT_j380 HLT_j400_a10r_L1J100 HLT_j400_a10_lcw_L1J100 HLT_j260_320eta490	HLT_3j175 HLT_4j100 HLT_5j70_L14J15 HLT_6j60 HLT_6j45_0eta240 HLT_7j45 HLT_10j40_L14J20	HLT_xe90_mht_L1XE50	HLT_ht1000_L1J10
period D4-E3	HLT_j380 HLT_j400_a10_lcw_L1J100 HLT_j400_a10r_L1J100 HLT_j260_320eta490	HLT_3j175 HLT_4j100 HLT_5j70_L14J15 HLT_5j65_0eta240_L14J15 HLT_6j60 HLT_6j45_0eta240 HLT_7j45_L14J15 HLT_10j40_L14J15	HLT_xe110_mht_L1XE50 (*)	HLT_ht1000_L1J100
period F1	HLT_j380 HLT_j420_a10_lcw_L1J100 HLT_j420_a10r_L1J100 HLT_j260_320eta490	HLT_3j200 HLT_4j100 HLT_5j70_L14J15 HLT_5j65_0eta240_L14J15 HLT_6j60 HLT_6j45_0eta240 HLT_7j45_L14J15 HLT_10j40_L14J15	HLT_xe110_mht_L1XE50 (*)	HLT_ht1000_L1J100
period	HLT_j380	HLT_3j200	HLT_xe110_mht_L1XE50	HLT_ht1000_L1J100

Trigger Group	Single jet	Multi jets	MET	HT
F2-	HLT_j420_a10_lcw_L1J100 HLT_j420_a10r_L1J100 HLT_j260_320eta490	HLT_4j100 HLT_5j70_L14J15 HLT_5j65_0eta240_L14J15 HLT_6j60 HLT_6j45_0eta240 HLT_7j45_L14J15 HLT_10j40_L14J15		

(a) Analyses using MET triggers should also check events that fire **HLT_noalg_L1J400** to see if any of these events were **not** fired by the met trigger in the analysis phase-space.

(*) HLT_xe100_mht_L1XE50 ran unprescaled during D4-F1 other than at the start of some high-luminosity runs.

Bjet

Trigger Group	1 Bjet	2 Bjet	Bjet + muon
period A	HLT_j175_bmv2c2040_split HLT_j65_bmv2c2070_split_3j65_L14J15 HLT_j55_bmv2c2060_split_ht500_L14J15	HLT_j150_bmv2c2077_split_j50_bmv2c2077_split HLT_2j35_bmv2c2070_split_2j35_L14J15 HLT_2j65_bmv2c2070_split_j65 HLT_2j55_bmv2c2070_split_ht300_L14J15 HLT_j100_2j55_bmv2c2077_split	
period B-D3	HLT_j175_bmv2c2040_split HLT_j65_bmv2c2070_split_3j65_L14J15 HLT_j55_bmv2c2060_split_ht500_L14J15	HLT_j150_bmv2c2060_split_j50_bmv2c2060_split HLT_2j35_bmv2c2060_split_2j35_L14J15.0ETA25 HLT_2j35_bmv2c2050_split_2j35_L14J15 HLT_2j70_bmv2c2070_split_j70 HLT_2j55_bmv2c2060_split_ht300_L14J15 HLT_j100_2j55_bmv2c2060_split	
period D4-	HLT_j175_bmv2c2040_split HLT_j225_bmv2c2060_split HLT_j275_bmv2c2070_split HLT_j300_bmv2c2077_split HLT_j360_bmv2c2085_split HLT_j55_bmv2c2060_split_ht500_L14J15 HLT_j75_bmv2c2070_split_3j75_L14J15.0ETA25	HLT_2j35_bmv2c2060_split_2j35_L14J15.0ETA25 HLT_2j45_bmv2c2077_split_3j45 (1) HLT_2j45_bmv2c2077_split_3j45_L14J15.0ETA25 HLT_2j55_bmv2c2060_split_ht300_L14J15 HLT_2j70_bmv2c2060_split_j70 HLT_2j75_bmv2c2070_split_j75 HLT_j100_2j55_bmv2c2060_split HLT_j150_bmv2c2060_split_j50_bmv2c2060_split HLT_j175_bmv2c2060_split_j50_bmv2c2050_split	

- Triggers listed from a period D4 were active for a whole year

(1) Seeded from L15J15

VBF/VH

Trigger Group	photon+VBF	e/mu+VBF
period A	HLT_g25_medium_L1EM22VHI_4j35_0eta490_invm700 HLT_g75_tight_3j50noL1_L1EM22VHI	HLT_e15_lhth HLT_m

Trigger Group	photon+VBF	e/mu+VBF
		HLT_mu6_ivarloose_2j40_0eta490_invm1000_I
period B-	HLT_g25_medium_L1EM22VHI_4j35_0eta490_invm700 HLT_g75_tight_3j50noL1_L1EM22VHI	HLT_e15_lht HLT_m HLT_mu6_ivarloose_2j40_0eta490_invm1000_I

Special physics triggers

Trigger Group	LLP cal-ratio	LLP muvtx	razor	
period A	HLT_j30_jes_PS_llp_L1TAU60 HLT_j30_jes_PS_llp_noiso_L1TAU60	HLT_j30_muvtx HLT_j30_muvtx_noiso	HLT_j30_xe10_razor170	HLT_g10_loose_I
period B-	HLT_j30_jes_PS_llp_L1TAU60 HLT_j30_jes_PS_llp_noiso_L1TAU60	HLT_j30_muvtx HLT_j30_muvtx_noiso	HLT_j30_xe10_razor185	HLT_g10_loose_I

* Introduced ExoDelayed stream for dijet TLA chains from period D

Triggers in 2015

Introduction

- Shown in **bold** are the triggers that are in the baseline menu and supposed not to be pre-scaled throughout 2015.

Data Periods

Some useful COMA ConditionsMetadata links for data15_13TeV are:

- COMA Periods of data15_13TeV [↗](#)
- COMA Period Runs in data15_13TeV [↗](#)
- COMA Period Triggers in data15_13TeV [↗](#)
- COMA Period Streams in data15_13TeV [↗](#)

Some important additional Period-specific notes:

- **period C1 (toroid off)** - high lumi baseline menu used, to allow lots of rate for muon calibration.
- **period C2 (toroid on)** - lumi ~ 3E32 (110 coll. bunches , runs 270949, 270953, 271048)
- **period C3 (toroid on)** - lumi ~ 7E32 (254 coll. bunches, runs 271298, 271421)
- **period C4 (toroid on)** - lumi ~ 1.5E33 (414 coll. bunches, runs 271516, 271595)
- **period D1,D2 toroid off** - high lumi menu (3E33) used, focus on muon calibration - lumi 1.5-2 E32 , ~74 coll. bunches, run 276161, 276176, 276181, 276183 - lumi ~ 3.6-4.6 E32 , 146 coll. bunches, runs 276189, 276212, 276245
- **period D3 toroid on** - lumi ~ 4.6E32, 146 coll bunches, run 276262 , lumi ~ 5-6E32, 206 coll bunches, run 276329 ,276330, 276336
- **period D4 toroid on** - lumi ~ 7-8E32, 302 coll bunches, run 276416, 276511, 276689
- **period D5 toroid on** - lumi ~ 1E33, 254 coll bunches, run 276731
- **period D6 toroid on** - lumi ~ 9E32-1E33, 447 coll bunches, run 276778, 276790, 276952
- **period E** - lumi up to ~ 3E33 (1176 bunches)
- **period F** - lumi up to ~ 3.2E33
- **period G** - lumi up to ~ 3.8E33

- **period H** - lumi up to ~ 4.6E33
- **period I** - lumi up to ~ 1E31 (ALFA)
- **period J** - lumi up to ~ 4.5E33

Note: Due to Pixel/IBL restrictions, L1 output rate is severely limited in period C +D. Therefore a higher luminosity point menu wrt the actual peak lumi of a run must be used, to control the rate.

Rulebook tags

We started to tag the rulebook package in order to be able to trace chain rules when a new AtlasP1HLT cache is deployed and used online. Please, find below the tags corresponding to different releases and periods together with the corresponding JIRAs.

- Tag for AtlasP1HLT, 20.2.1.5 (period C1): TrigMenuRulebook-00-01-42 [↗](#) ATR-9923 [↗](#)
- Tag for AtlasP1HLT, 20.2.1.5 (period C2-C4): TrigMenuRulebook-00-01-43 [↗](#) to TrigMenuRulebook-00-01-58 [↗](#) ATR-11652 [↗](#)
- Tag for AtlasP1HLT, 20.2.1.7 (period D1-D5) ATR-11983 [↗](#) , rules ATR-11756 [↗](#), TrigMenuRulebook-00-01-74 [↗](#) to TrigMenuRulebook-00-01-91 [↗](#)
- Tag for AtlasP1HLT, 20.2.1.8 (period D6) ATR-12068 [↗](#) , rules ATR-11756 [↗](#), TrigMenuRulebook-00-01-98 [↗](#)
- Tag for AtlasP1HLT, 20.2.1.8-20.2.3.2 (period E) ATR-12068 [↗](#) ATR-12193 [↗](#) ATR-12250 [↗](#), TrigMenuRulebook-00-02-06 [↗](#) to TrigMenuRulebook-00-02-23 [↗](#)
- Tag for AtlasP1HLT, 20.2.3.2 - 20.2.3.3 (period F) ATR-12250 [↗](#) ATR-12383 [↗](#), TrigMenuRulebook-00-02-23 [↗](#) to TrigMenuRulebook-00-02-31 [↗](#)
- Tag for AtlasP1HLT, 20.2.3.3 - 20.2.3.4 (period G) ATR-12383 [↗](#) ATR-12439 [↗](#), TrigMenuRulebook-00-02-31 [↗](#) to TrigMenuRulebook-00-02-36 [↗](#)
- Tag for AtlasP1HLT, 20.2.3.4 - 20.2.3.5 (period H) ATR-12439 [↗](#) ATR-12489 [↗](#), TrigMenuRulebook-00-02-36 [↗](#) to TrigMenuRulebook-00-02-38 [↗](#)
- Tag for AtlasP1HLT, 20.2.3.5 - 20.2.3.6 (period I) ATR-12489 [↗](#) ATR-12549 [↗](#), Keys were mostly tuned by hand based on observed rates during the 90m runs TrigMenuRulebook-00-02-42 [↗](#) to TrigMenuRulebook-00-02-30 [↗](#)
- Tag for AtlasP1HLT, 20.2.3.6 - 20.2.3.7 (period J) ATR-12549 [↗](#), TrigMenuRulebook-00-02-51 [↗](#) to TrigMenuRulebook-00-02-64 [↗](#)

Egamma (+MET)

Trigger Group	Single electron	Multi electron	Single photon	
period C2-C4,D3-D6, E, F, G, H, J	HLT_e24_lhmedium_L1EM20VH() HLT_e60_lhmedium HLT_e120_lhloose	HLT_2e12_lhvloose_L12EM10VH HLT_e17_lhloose_2e9_lhloose	HLT_g120_loose HLT_g200_etcut	HLT_g3 HLT_g3 HLT_2g HLT_3g

Note:

- In period C several variants for electron triggers (nod0/nodphires/nodeta, loose/vloose/lhloose, etc...) are also present in the menu and unprescaled. Scale factors / efficiency only provided for nominal LH trigger.
- (*) The lowest unprescaled single electron is e24_lhmedium_L1EM18VH. The previous candidate HLT_e24_lhmedium_L1EM18VH is replaced by HLT_e24_lhmedium_iloose_L1EM18VH in period C3-C4. In period E, sometimes the lumi key for 4.5E33 is used at the start of runs, which means effectively the first

5-10' of run have L1_EM18VH prescaled, and the unprescaled L1 item is L1_EM20VH.

- (**) For 2photon item, in period E sometimes lumi key for 4.5E33 is used at start, which means effectively first 5-10' of run have L1_2EM15 prescaled. HLT_g35_loose_g25_loose seeded by L1_2EM15VH remained unprescaled during the run.

Muon

Trigger Group	Single muon	Multi muon
period C2-C4, D3-D6, E, F, G, H, J	HLT_mu20_iloose_L1MU15 HLT_mu40 HLT_mu60_0eta105_monly	HLT_2mu10 HLT_3mu6 HLT_3mu6_monly HLT_mu18_2mu4noL1 HLT_mu18_mu8noL1

Electron + Muon

Trigger Group	e+mu	multi-leptons
period C2-C4, D3-D6	HLT_e17_lhloose_mu14 HLT_e7_lhmedium_mu24	HLT_2e12_lhloose_mu10 HLT_e12_lhloose_2mu10

Muon + Photon

Trigger Group	muon+multi-photon	multi-muon+photon
period C2-C4, D3-D6	HLT_g25_medium_mu24 HLT_2g10_loose_mu20	HLT_g15_loose_2mu10_monly

Bphysics

Trigger Group	Bmumu	Jpsimumu	Upsimumu	B
period C2-C4, D3-D6	HLT_2mu4_bBmumu	HLT_2mu4_bJpsimumu HLT_mu18_2mu0noL1_JpsimumuFS(*)	HLT_2mu4_bUpsimumu	HLT_2mu4_bBmumu HLT_2mu4_bBmumu HLT_2mu4_bBmumu

Note:

- (*) HLT_mu18_2mu0noL1_JpsimumuFS is originally designed for J/Psi T&P.
- 2MU4 prescaled out for a few lumi blocks at start of run 276952, and some other runs in period D (eg run 276511). it is off in run 276689.
- 2MU4 prescaled to 1kHz for lumis higher than 4E33. MU6_2MU4, is next threshold.

Taus (+ electron, muon, MET)

Trigger Group	Single tau	di-tau
period C2-C4,	HLT_tau80_medium1_tracktwo (TAU40)	HLT_tau35_loose1_tracktwo_tau25_loose1_tracktwo_L1TAU20

Trigger Group	Single tau	di-tau
D3-D6		
Full Year	HLT_tau80_medium1_tracktwo_L1TAU60	HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo_L1T

Note:

- (*) HLT_tau35_loose1_tracktwo_tau25_loose1_tracktwo_L1TAU20IM_2TAU12IM gets PS-1 in run 271595 (lumi start > 1.5E33), lowest unprescaled is HLT_tau35_medium1_tracktwo_tau25_medium1_tracktwo_L1TAU20IM_2TAU12IM or HLT_tau35_loose1_tracktwo_tau25_loose1_tracktwo (starts with jets at L1). This happens in several runs in period D3-D6, at least at start.

Jets, MET, Jet+MET, HT

Trigger Group	Single jet	Multi jets	MET	jet+MET	Sum Et	HT
period C2-C4, D3-D6, E, F, G, H, J	HLT_j260_320eta490 HLT_j360 HLT_j360_a10_lcw_sub_L1J100 (****) HLT_j360_a10r_L1J100 (****)(****) HLT_j360_a10_sub_L1J100 (****)	HLT_3j175 HLT_4j85 HLT_5j60 HLT_6j45	HLT_xe70_mht (**)	HLT_j80_xe80		HLT_ht850_L1J75 (*)

Note:

(*) HLT_ht850_L1J100 is the lowest unprescaled for the whole run, in run 271595 ($L > 1.5E33$) and run 276689; collecting roughly 2% more integrated luminosity.

(**) HLT_xe70_tc_lcw and HLT_xe70 were also unprescaled. Unprescaled variants of these triggers also exist with muon corrections, which may be preferable in certain contexts.

(***) HLT_j360_a10r_L1J100 had a bug in 2015, and is not recommended for usage

(****) Large-R jet primaries HLT_j360_a10_lcw_sub_L1J100, HLT_j360_a10r_L1J100 and HLT_j360_a10_sub_L1J100 have been removed from MC after a menu slimming. See a full list of removed chains in [ATR-16556](#).

Analyses using MET triggers should also check events that fire **HLT_noalg_L1J400** to see if any of these events were **not** fired by the met trigger in the analysis phase-space.

Bjet

Trigger Group	1 Bjet	2 Bjet	Bjet +
full year	HLT_j225_bloose HLT_j65_btight_3j65_L13J25.0ETA23 HLT_j70_bmedium_3j70_L13J25.0ETA23	HLT_2j35_btight_2j35_L13J25.0ETA23 HLT_2j45_bmedium_2j45_L13J25.0ETA23 HLT_j100_2j55_bmedium HLT_j150_bmedium_j50_bmedium	HLT_mu4_j40_dr0 HLT_mu4_4j40_dr0 HLT_mu4_3j45_dr0 various mu+jet+dr0

Trigger Group	1 Bjet	2 Bjet	Bjet +
		HLT_2j65_btight_j65 HLT_2j70_bmedium_j70	(*)
period C2-C4, D3-D6	HLT_j175_bmedium plus above	see above	

Note:

(*) In period C several variants for mu+Njet+dr05 triggers (starting with L1 Jets only, or L1 muon + jets) are present in the menu and unprescaled. Consult with b-jet convenors, or check SMK for a run, to see which one best suites your purpose. mu+Njet+dr05 were fully functional only in period C4.

(*) in period D the mu+jet+dr (w oe w/o bperf) are fixed and better.

VBF

Trigger Group	photon+VBF	mu+V
period C2-C4, D3-D6	HLT_g20_loose_L1EM18VH_2j40_0eta490_3j25_0eta490_invm700	HLT_mu6_2j40_0eta490_invm400_L10

Note:

(*) lowest unprescaled in run 271595 (L>1.5E33) and 276689 is

HLT_mu6_2j40_0eta490_invm600_L1MU6_J30.0ETA49_2J20.0ETA49. lowest unprescaled in run 271595 (L>1.5E33) is HLT_2j55_bmedium_2j55

Special physics triggers

Trigger Group	LLP cal-ratio	LLP muvtx	mono pole	tau 3mu, l
period C2-C4, D3-D6	HLT_j30_jes_PS_llp_L1TAU30 (*)	HLT_j30_muvtx_noiso	g0_hiptrt_L1EM18VH	HLT_mu11_2mu4noL1_nsc (x) HLT_mu20_mu6noL1_nsc HLT_mu20_msonly_mu6n

Note:

(*) HLT_j30_jes_PS_llp_L1TAU30, TAU40 get PS=-1 from run 271595 due to CPU constraints. Continues in D3-D6. Lowest unprescaled is HLT_j30_jes_PS_llp_L1TAU60

(x) l2msonly version of these gets PS=-1 from run 271595 due to CPU constraints. Stays off in period D.

Triggers in Run-1 (2012,2011,2010)

- Lowest un-prescaled triggers in 2012
- Lowest un-prescaled triggers in 2011
- Lowest un-prescaled triggers in 2010

Major updates:

-- ArantxaRuizMartinez - 3-May-2018 -- ImmaRiu - 1-May-2012 -- OlyaIgonkina - 15-Sep-2011 -- AnnaSfyrla - 27-Sep-2010

Responsible: ArantxaRuizMartinez, JavierMontejo

Last reviewed by: **Never reviewed**

- Menu1E33-1.2E34-Final2016.pdf: Tables containing lowest unprescaled combined and special triggers (pages 8-10)

This topic: Atlas > LowestUnprescaled

Topic revision: r341 - 2026-07-23 - StanLai



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```
var _paq = window._paq = window._paq || []; /* tracker methods like "setCustomDimension" should be
called before "trackPageView" */ _paq.push(['trackPageView']); _paq.push(['enableLinkTracking']);
(function() { var u="https://webanalytics.web.cern.ch/"; _paq.push(['setTrackerUrl', u+'matomo.php']);
_paq.push(['setSiteId', '5']); var d=document, g=d.createElement('script'),
s=d.getElementsByTagName('script')[0]; g.async=true; g.src=u+'matomo.js'; s.parentNode.insertBefore(g,s);
})();
```